Beyond "Randomized": Designing and Evaluating Randomization in Clinical Trials

Stefanie Schoenen (RWTH Aachen)
Daniel Bodden (RWTH Aachen)
Franz König (Medical University of Vienna)
Ralf-Dieter Hilgers (Sigmund Freud Private University Vienna)

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Contact: stschoenen@ukaachen.de · daniel.bodden@rwth-aachen.de

MEDIZINISCHE UNIVERSITÄT WIEN
SIGMUND FREUD PRIVATE UNIVERSITY VIENNA
RWTH AACHEN UNIVERSITY

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Realise RDI

compRehensive mEthodological and operational Approach to cLinical trIalS in ultra-rarE Diseases

COCIR Advancing Healthcare · efpia · EuropaBio The European Association for Bioindustries · MedTech Europe from diagnosis to cure · Vaccines Europe · Co-funded by the European Union

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Workshop agenda

Time	Session	Topics
8:30-8:40 (10 min)	**Welcome & workshop overview**	• Overview of the workshop topics and flow
8:40-9:40 (60 min)	**I** **Block I — General introduction to randomization**	• History and evolution of randomization • Randomization, allocation concealment, and blinding • Randomization procedures and their key characteristics • Regulatory perspectives on randomization
9:40-10:30 (50 min)	**II** **Block II — Bias & transparent reporting**	• (Allocation) Bias in randomized clinical trials, • Transparent reporting: what to report in the trial protocol and publication
10:30-11:00 (30 min)	**Coffee Break**	
11:00-12:15 (75 min)	**III** **Block III — Randomization in complex trial designs**	• Randomization in small-sample group-sequential designs • (Stratified) randomization and allocation bias in multi-endpoint trials • Randomization in multi-arm and platform trials
12:15-12:30 (15 min)	**IV** **Block IV — Recommendations & take-home messages**	• Key lessons learned and further topics • Open-research question and working groups

Block I: General introduction to randomization

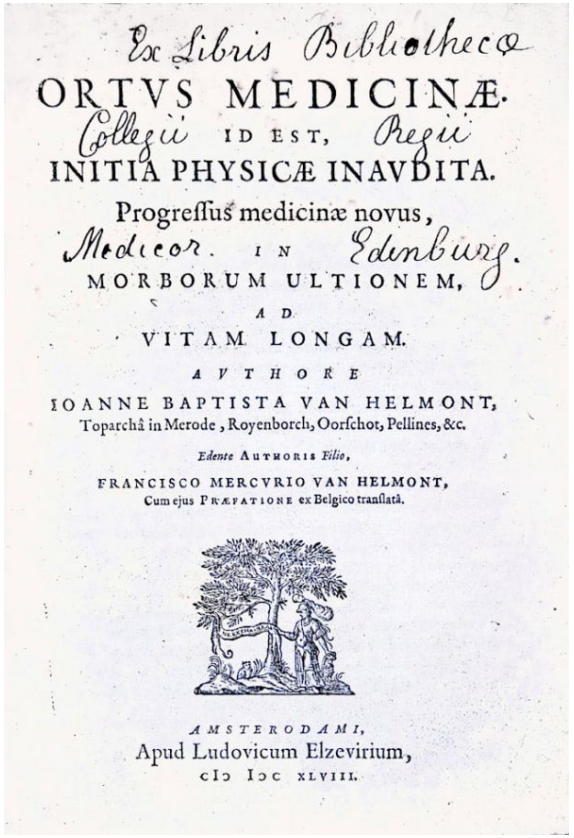
- History and evolution of randomization
- Randomization, allocation concealment, and blinding
- Randomization procedures and their key characteristics
- Regulatory perspectives on randomization



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History of randomization: Randomization by lots

1662



Ex Libris Bibliotheco

ORTVS MEDICINÆ.

Collegii ID EST, Regii

INITIA PHYSICÆ INAVDITA.

Progressus medicinæ novus,

Medicor. IN Edinburg.

MORBORUM ULTIONEM,

AD

VITAM LONGAM.

AVTHORE

IOANNE BAPTISTA VAN HELMONT,

Toparchâ in Merode, Royenborch, Oorschot, Pellines, &c.

Edente AUTHORIS Filio,

FRANCISCO MERCVRIO VAN HELMONT,

Cum ejus PRÆFATIONE ex Belgico translatâ.

AMSTERODAMI,

Apud Ludovicum Elzevirium,

cIↃ IↃC XLVIII.

"Let us take … 200, or 500 poor people, that have fevers, pleurisies … Let us divide them in halves, let us cast lots … I will cure them without bloodletting … we shall see how many funerals both of us shall have. "

Jan Baptist van Helmont, Oriatrike (English ed., 1662)

History of randomization: Ronald Fisher

1935


The
Design of Experiments

By

R. A. Fisher, Sc.D., F.R.S.

Oliver and Boyd
Edinburgh: Tweeddale Court
London: 33 Paternoster Row, E.C.
1935

- In 1935 Ronald Fisher published his book “The Design of Experiments“
- Laid the methodological foundation for randomization

> *[…] the physical act of randomization […] is necessary for the validity of any test of significance […]*

Fisher RA (1935). The Design of Experiments. p. 51

> *[…] This does not mean that the experimenter writes down the names of the varieties...in any order that may occur to him, but that he carries out a physical experimental process of randomization […]*

Fisher RA (1935). The Design of Experiments. p. 56

History of randomization: Austin Bradford Hill

1937

- Sir Austin Bradford Hill published his book “Principles of Medical Statistics”
- Promotes the use of alternating sequences as trial design

The Problems of Clinical Trials
With methods of treatment the main questions to be settled are usually these :
(a) How can the patients be effectively allocated to the two groups which are to be compared-which we can refer to as the treated and control groups.
(b) What criterion or criteria can be used as evidence of the effects of treatment.
(c) On how many patients will the trials have to be made to give reliable results.

Sir Bradford A Hill (1938). Principles of Medical Statistics

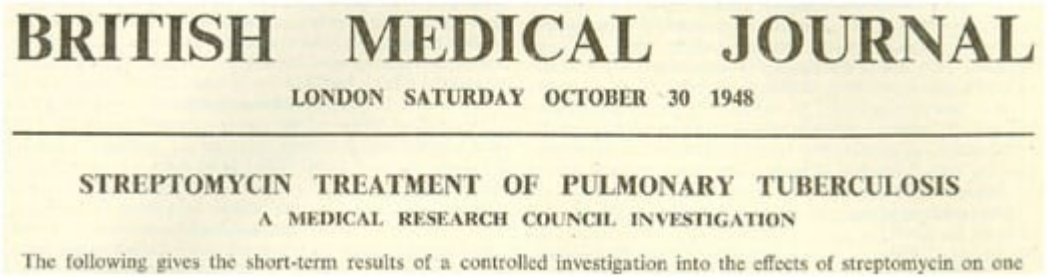
(a) ALLOCATION TO GROUPS
By the allocation of patients to the two groups we want to ensure that these two groups are alike except in treatment […] with reasonably large numbers, by a random division of the patients, the first being given treatment A, the second being orthodoxly treated and serving as a control […]

Sir Bradford A Hill (1938). Principles of Medical Statistics

History of randomization: Trial of Streptomycin in pulmonary tuberculosis

1946

- First RCT of streptomycin in pulmonary tuberculosis
- Responsible statistician: Bradford Hill



BRITISH MEDICAL JOURNAL

LONDON SATURDAY OCTOBER 30 1948

STREPTOMYCIN TREATMENT OF PULMONARY TUBERCULOSIS

A MEDICAL RESEARCH COUNCIL INVESTIGATION

The following gives the short-term results of a controlled investigation into the effects of streptomycin on one

Results:

- Streptomycin & bed rest group: 4/55 patients died before end of six months
- Bed rest group: 14/52 patients died before end of six months

Rationales of randomization

1948

“ The random allocation has not only removed personal responsibility from the clinician and possible bias in his process of choosing patients, but has on the whole effectively equated the groups – fundamental, of course to the general comparisons. “ (p. 792)

British Medical Journal Editorial Board. The controlled therapeutic trial. Br Med J. 1948;2:791-792.

Early literature mentioned three rationales for randomization:

i. unpredictability in treatment assignments-> mitigating selection bias
ii. similarity in treatment groups of known and unknown confounders
iii. randomization itself provides a basis of inference

Rosenberger WF, Uschner D, Wang Y. Randomization: The forgotten component of the randomized clinical trial. Stat Med. 2019;38(1):1–30

Let us talk about bias in randomized clinical trials

- **Allocation Bias:** A type of selection bias that occurs when future treatment allocations can be predicted from previous ones, usually due to improper blinding

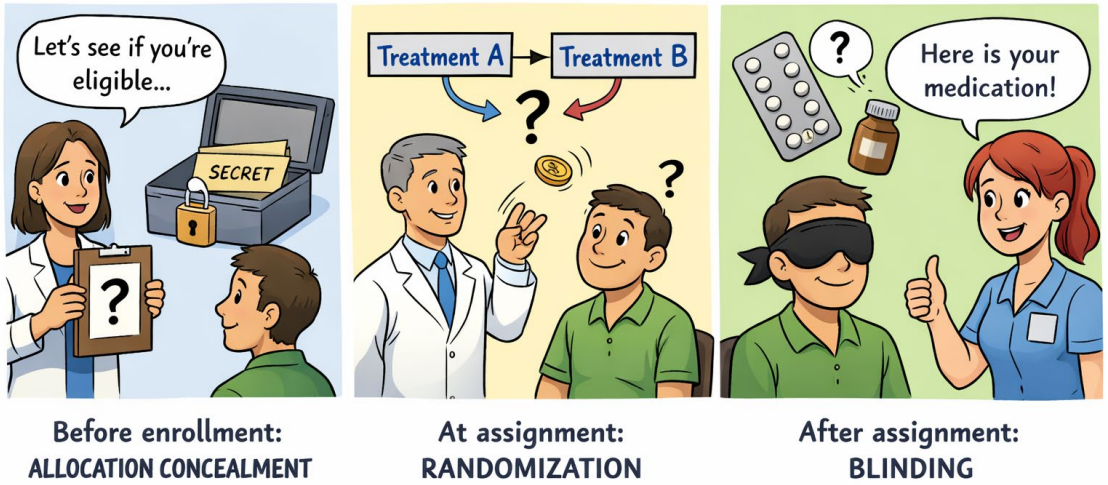
- **Performance Bias:** Systematic differences in care provided apart from the intervention being evaluated, e.g., due to prejudiced staff

- **Detection Bias**: Systematic differences between comparison groups in how outcomes are ascertained, diagnosed or verified

- **Attrition Bias:** Systematic differences between comparison groups in withdrawals or exclusions of participants from the results of a study.

https://catalogofbias.org/

Randomization, allocation concealment, and blinding are not the same



- **Allocation concealment:** prevents foreknowledge of the next assignment before enrolment
- **Randomization:** creates an unpredictable treatment assignment mechanism
- **Blinding:** prevents knowledge of assigned treatment after allocation

Mechanism to prevent bias

- **Allocation Bias:** A type of selection bias that occurs when future treatment allocations can be predicted from previous ones, usually due to improper blinding

randomization, blinding, allocation concealment

- **Performance Bias:** Systematic differences in care provided apart from the intervention being evaluated, e.g., due to prejudiced staff

standardization of care and blinding

- **Detection Bias**: Systematic differences between comparison groups in how outcomes are ascertained, diagnosed or verified

blinding of outcome assessors and patients

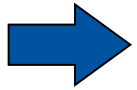
- **Attrition Bias:** Systematic differences between comparison groups in withdrawals or exclusions of participants from the results of a study.

blinding, intention to treat analysis

https://catalogofbias.org/

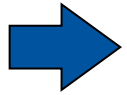
How can we get rid of bias?

- **Pre-assignment of treatment:**
  - randomization
  - allocation concealment



can always be implemented

- **Post-assignment of treatment:**
  - blinding

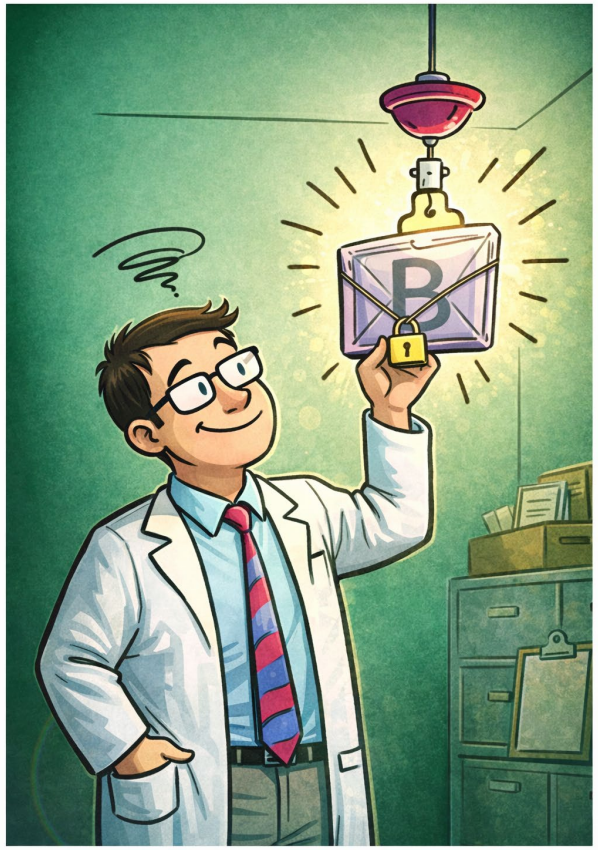


can not always be implemented

Blinding at any price



Why not “open“ studies?



Deciphering the allocation concealment scheme

Concealment:

Compared with adequately concealed trials, trials with **inadequate or unclear concealment overestimated treatment effects**: ROR 0.92 (95% CI 0.88–0.97). Where concealment was only unclear: ROR 0.85 (95% 0.74–0.98).

Blinding:

Lack of **double blinding*** probably leads to an effect overestimation for **subjective outcomes** with moderate certainty (ROR 0.70, 95% CI 0.53–0.89), and **any outcomes** with low certainty (ROR 0.89, 95% CI 0.77–1.04).

ROR = ratio of odds ratios; ROR < 1 means the trials lacking the safeguard reported the larger effect.
* as reported by authors

Wang Y, Parpia S, Couban R, et al. Compelling evidence from meta-epidemiological studies demonstrates overestimation of effects in randomized trials that fail to optimize randomization and blind patients and outcome assessors. J Clin Epidemiol. 2024;165:111211. doi:10.1016/j.jclinepi.2023.11.001

How to allocate patients?



Original Contribution FREE

Effect of a Monetary Sanction on Immunization Rates of Recipients of Aid to Families With Dependent Children

Larry C. Kerpelman, PhD; David B. Connell, PhD; Walter J. Gunn, PhD

Cite Permissions Metrics

JAMA
Published Online: July 5, 2000
2000;284;(1):53-59. doi:10.1001/jama.284.1.53

“To evaluate the effect of the sanction, a randomized controlled trial was instituted in Muscogee County, Georgia […]“

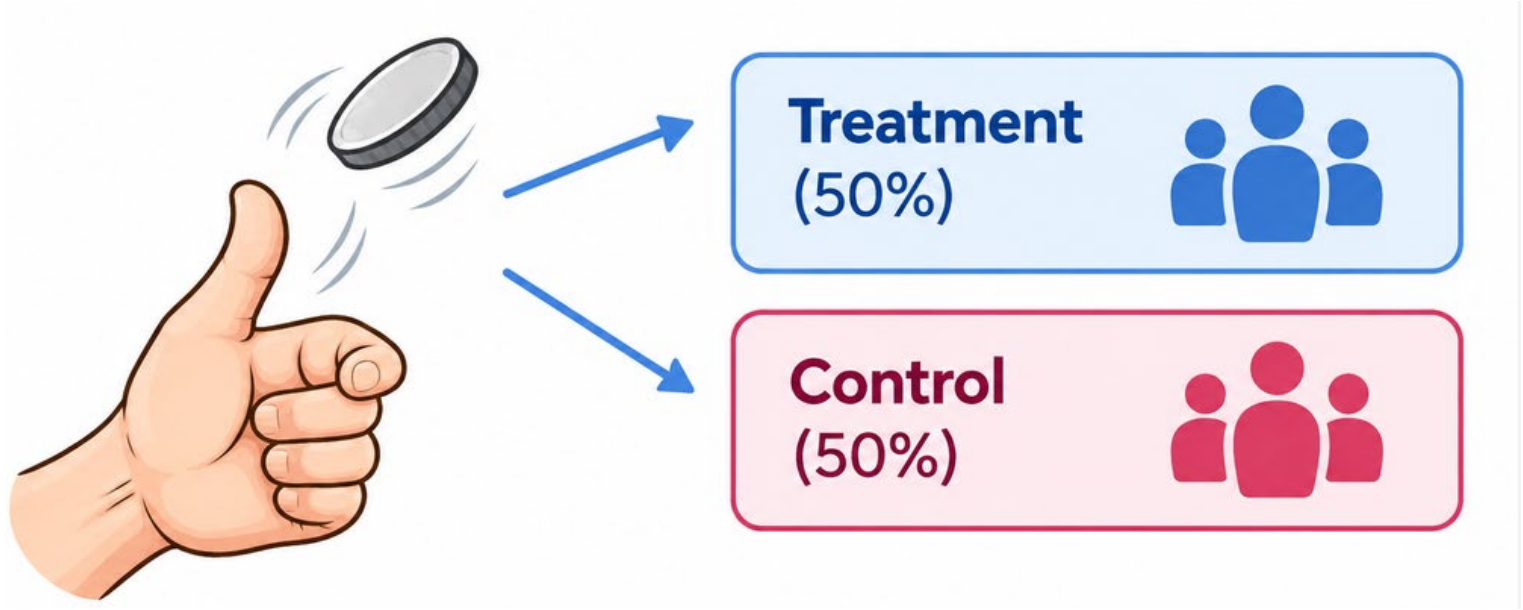
“From those families, every third family was assigned to the control group until 1000 cases were selected. Every third family from the remaining 2600 was assigned to the intervention group until 1500 cases were selected“

Deterministic allocation rule!

Kerpelman LC, Connell DB, Gunn WJ. Effect of a Monetary Sanction on Immunization Rates of Recipients of Aid to Families With Dependent Children. *JAMA*. 2000;284(1):53–59. doi:10.1001/jama.284.1.53

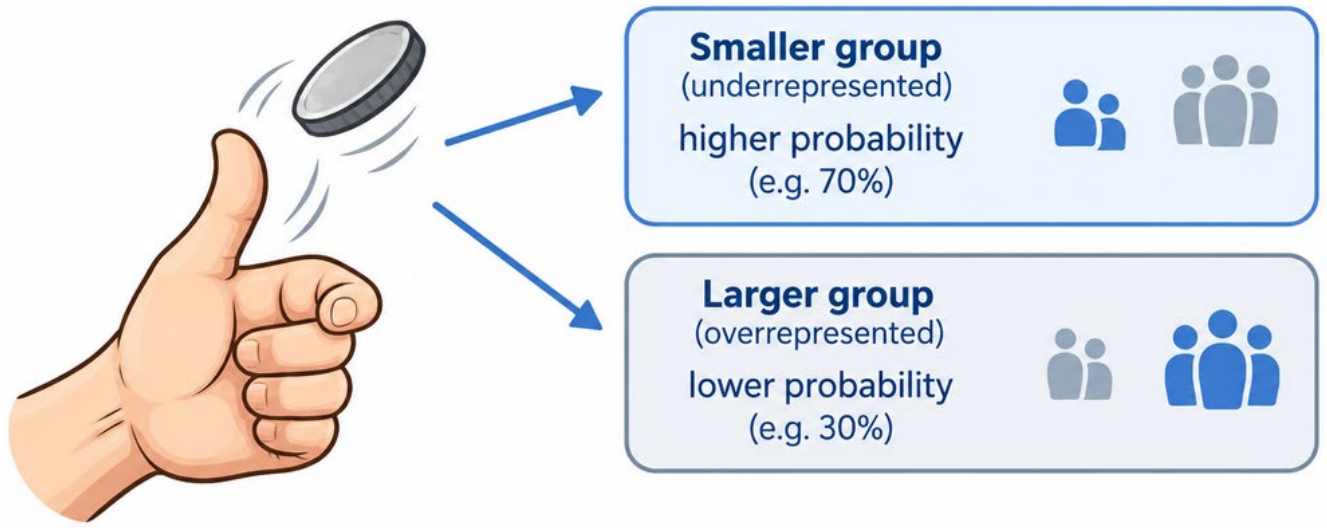
Complete Randomization (CR)

- **Allocation by fair coin toss:** Participants (n) are randomly assigned to the treatment or control group with equal probability ($\frac{1}{2}$ each).



Efron Biased Coin Design (EBC(p))

- **Allocation by a biased coin toss:** Participants (n) are randomly assigned using a biased coin that favors the group with fewer participants with probability *p*. If group sizes are balanced, each group is assigned with probability 1/2.



Random Allocation Rule (RAR*)

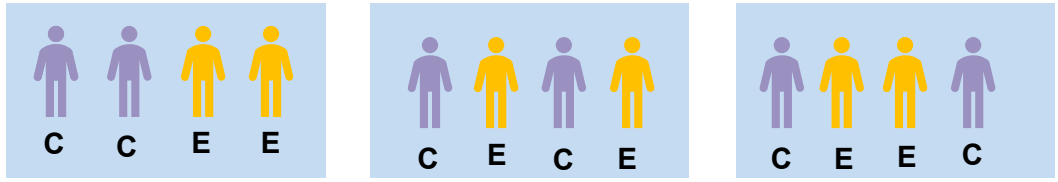
- Exactly n/2 patients are assigned to each group. This can be represented by an urn model containing n/2 treatment and n/2 control balls. A ball is drawn without replacement, and the corresponding treatment is assigned.

*The abbreviation RAR is often also used for response adaptive randomization

Permuted Block Randomization (PBR(a))

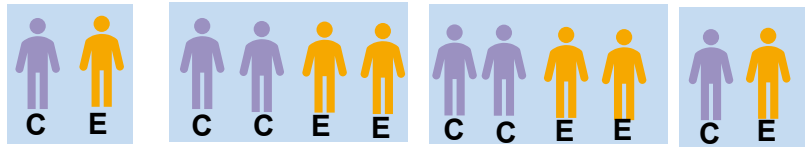
- Patients are randomized in blocks of size a; within each block, allocation follows Random Allocation Rule.

Example (n=12, a=4):



- **Extension-> Random block design:** In this case, the block length is randomly selected from a predefined set of possible block lengths rather than being fixed.

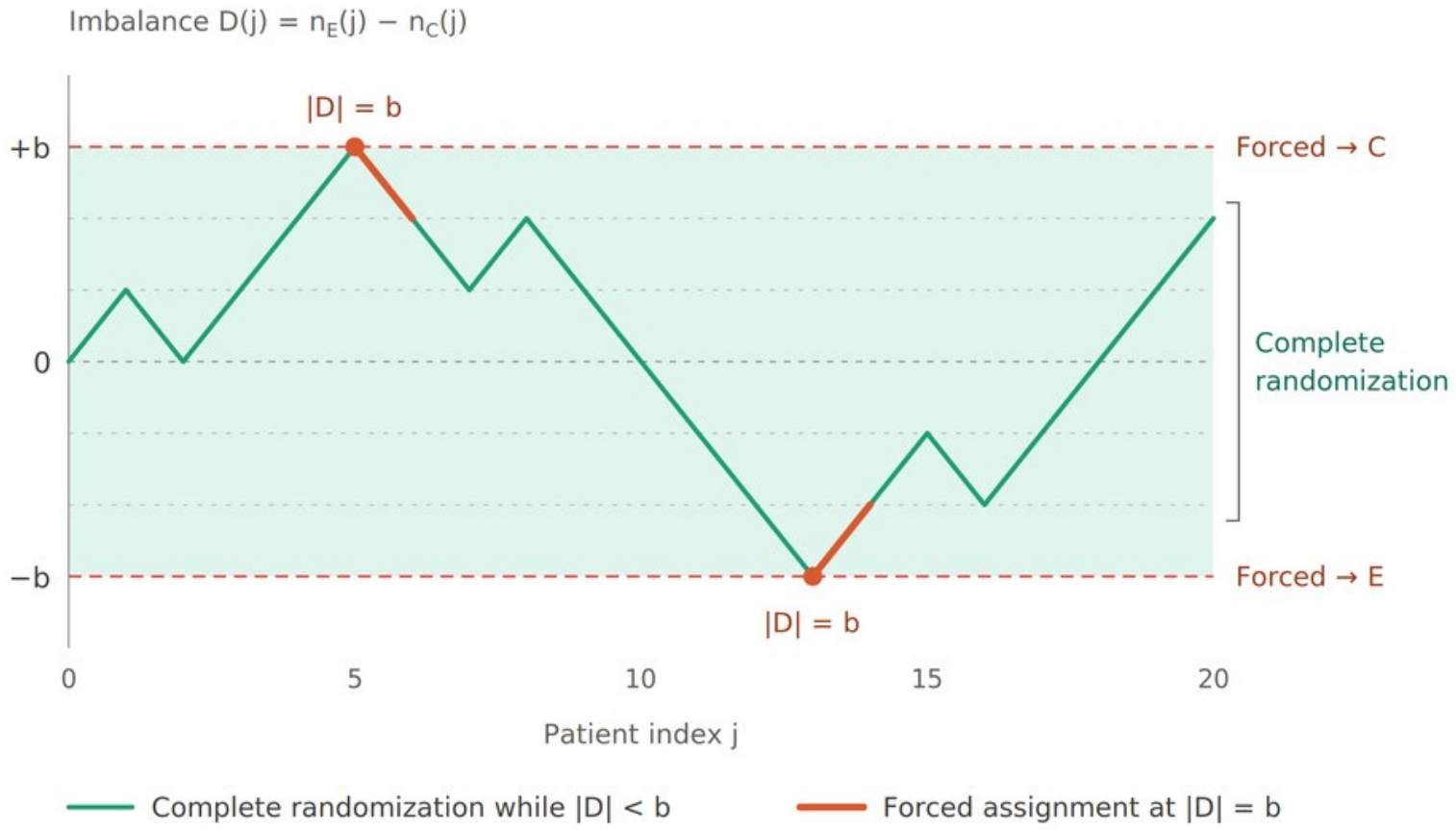
Example (n=12, a=2,4,6):



Randomization procedures

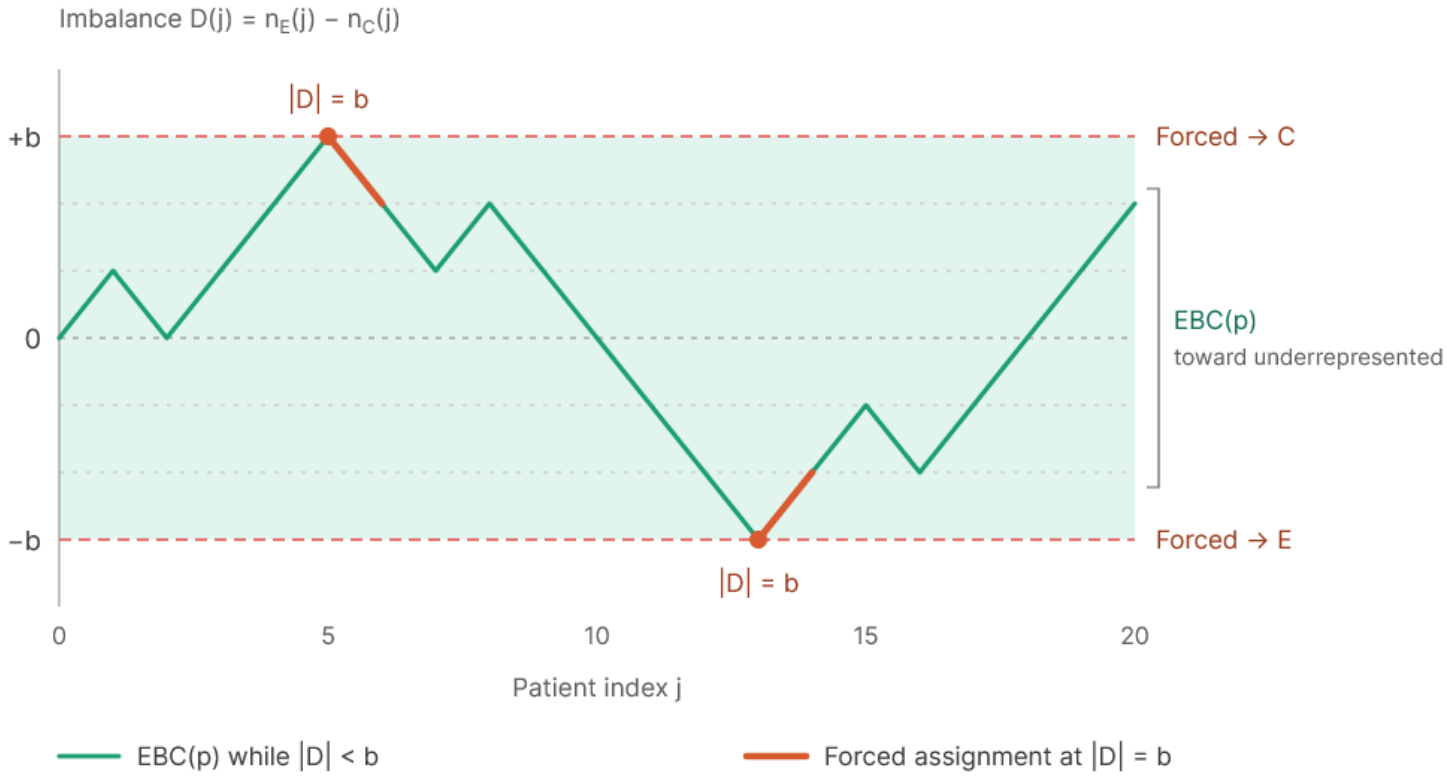
Big Stick Design (BSD(b))

- Complete Randomization until imbalance b is reached; thereafter, patients are deterministically assigned to the underrepresented group.

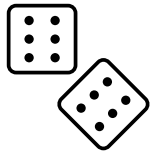


Chen's Urn Design (Chen(p,b))

- Hybrid of Efron's Biased Coin and Big Stick Design. Assignments follow EBC(p) while the imbalance is below the limit b; once b is reached, the next patient is deterministically assigned to the underrepresented group.



Beyond Randomness: Key characteristics of randomization procedures



Randomness

- Reduced predictability



Balance

- **Final balanced allocation**
  → Similar numbers of participants in each treatment group at trial completion
- **Balanced allocation within the trial**
  → Treatment groups remain approximately balanced throughout recruitment

Randomization procedure needs to find the trade-off between randomness/ reduced predictability and allocation balance.

Shiny tool

Open the app in your browser or smartphone:

https://danielbodden.de/randomizer

Properties of the randomization procedures

Abbr.	Procedure	Mechanism
CR	Complete Randomization	Allocation by a fair coin toss.
EBC(p)	Efron's Biased Coin	Biased coin toss (p) that favors the under-represented group.
RAR	Random Allocation Rule	Exactly n/2 patients are assigned to each group.
PBR(a)	Permuted Block Rand.	Patients are randomized in blocks of size a; within each block, allocation follows RAR.
rPBR(a)	Random permuted Block Rand.	The block length is randomly selected from a predefined set of possible block lengths rather than being fixed.
BSD(b)	Big Stick Design	CR until imbalance b is reached, then deterministic assignment to the under-represented group.
Chen(p,b)	Chen's Urn Design	Biased coin (p) that favors the under-represented group, allowing for imbalance within a limit b.

final balance

maximum tolerated imbalance

no forced balanced allocation

ICH E9

2.3.2 Randomisation

Randomisation introduces a deliberate element of chance into the assignment of treatments to subjects in a clinical trial. During subsequent analysis of the trial data, it provides a sound statistical basis for the quantitative evaluation of the evidence relating to treatment effects. It also tends to produce treatment groups in which the distributions of prognostic factors, known and unknown, are similar. In combination with blinding, randomisation helps to avoid possible bias in the selection and allocation of subjects arising from the predictability of treatment assignments.

The randomisation schedule of a clinical trial documents the random allocation of treatments to subjects. In the simplest situation it is a sequential list of treatments (or treatment sequences in a crossover trial) or corresponding codes by subject number. The logistics of some trials, such as those with a screening phase, may make matters more complicated, but the unique pre-planned assignment of treatment, or treatment sequence, to subject should be clear. Different trial designs will require different procedures for generating randomisation schedules. The randomisation schedule should be reproducible (if the need arises).

Although unrestricted randomisation is an acceptable approach, some advantages can generally be gained by randomising subjects in blocks. This helps to increase the comparability of the treatment groups, particularly when subject characteristics may change over time, as a result, for example, of changes in recruitment policy. It also provides a better guarantee that the treatment groups will be of nearly equal size. In crossover trials it provides the means of obtaining balanced designs with their greater efficiency and easier interpretation. Care should be taken to choose block lengths that are sufficiently short to limit possible imbalance, but that are long enough to avoid predictability towards the end of the sequence in a block. Investigators and other relevant staff should generally be blind to the block length; the use of two or more block lengths, randomly selected for each block, can achieve the same purpose. (Theoretically, in a double-blind trial predictability does not matter, but the pharmacological effects of drugs may provide the opportunity for intelligent guesswork.)

In multicentre trials (see Glossary) the randomisation procedures should be organised centrally. It is advisable to have a separate random scheme for each centre, i.e. to stratify by centre or to allocate several whole blocks to each centre. More generally, stratification by important prognostic factors measured at baseline (e.g. severity of disease, age, sex, etc.) may sometimes be valuable in order to promote balanced allocation within strata; this has greater potential benefit in small trials. The use of more than two or three stratification factors is rarely necessary, is less successful at achieving balance and is logistically troublesome. The

© EMEA 2006

use of a dynamic allocation procedure (see below) may help to achieve balance across a number of stratification factors simultaneously provided the rest of the trial procedures can be adjusted to accommodate an approach of this type. Factors on which randomisation has been stratified should be accounted for later in the analysis.

The next subject to be randomised into a trial should always receive the treatment corresponding to the next free number in the appropriate randomisation schedule (in the respective stratum, if randomisation is stratified). The appropriate number and associated treatment for the next subject should only be allocated when entry of that subject to the randomised part of the trial has been confirmed. Details of the randomisation that facilitate predictability (e.g. block length) should not be contained in the trial protocol. The randomisation schedule itself should be filed securely by the sponsor or an independent party in a manner that ensures that blindness is properly maintained throughout the trial. Access to the randomisation schedule during the trial should take into account the possibility that, in an emergency, the blind may have to be broken for any subject. The procedure to be followed, the necessary documentation, and the subsequent treatment and assessment of the subject should all be described in the protocol.

Dynamic allocation is an alternative procedure in which the allocation of treatment to a subject is influenced by the current balance of allocated treatments and, in a stratified trial, by the stratum to which the subject belongs and by balance within that stratum. Deterministic dynamic allocation procedures should be avoided. Some element of chance in the randomisation should be incorporated for each treatment allocation. Every effort should be made to retain the double-blind status of the trial. For example, knowledge of the allocation code may be restricted to a central trial office from where the dynamic allocation is controlled, generally through telephone contact. This in turn permits additional checks of eligibility criteria and establishes entry into the trial, features that can be valuable in certain types of multicentre trial. The usual system of pre-packing and labelling drug supplies for double-blind trials can then be followed, but the order of their use is no longer sequential. It is desirable to use appropriate computer algorithms to keep personnel at the central trial office blind to the treatment code. The complexity of the logistics and potential impact on the analysis should be carefully evaluated when considering dynamic allocation.

ICH E9 primarily refers to permuted block randomization when discussing randomization procedures.

Further regulatory view on randomization

Country	IDRAC	Document name	Randomization methodology mentioned				
			Unrest-ricted	Blocked	Covariate-adaptive	Response-adaptive	Cluster
China	285313	NMPA Announcement No. 2018/103: Publication of Statistical Guidelines for Bioequivalence Studies and Technical Guidelines for Bioequivalence Studies of Highly Variable Drugs, 17-Oct-2018 (English and Chinese Versions)		√			
China	228958	CFDA Announcement No. 2016/93: Announcement on Issuance of Technical Guidelines on Biostatistics of Drug Clinical Trials, 01-Jun-2016 (English and Chinese Versions)		√*	√		
EU	210331	EMA/CHMP/295050/2013 Guideline on Adjustment for Baseline Covariates in Clinical Trials, 26-Feb-2015		√	√		
EU	60165	CHMP Guideline CHMP/EWP/83561/2005: Clinical Trials in Small Populations, 27-Jul-2006				√	
EU	107822	CHMP/EWP/1343/01 Rev.1: Guideline on the Clinical Evaluation of Antifungal Agents for the Treatment and Prophylaxis of Invasive Fungal Disease, 22-Apr-2010		√			
Inter-national	16940	ICH Guideline Topic E9 Step 4: Statistical Principles for Clinical Trials, 05-Feb-1998	√	√*	√		
USA	180221	Good Review Practice: Clinical Review of Investigational New Drug Applications (Final), Dec-2013	√	√*	√	√	
USA	303151	Guidance for Industry: Adaptive Designs for Clinical Trials of Drugs and Biologics, Nov-2019			√	√	
USA	256451	Guidance for Industry: Antibacterial Therapies for Patients With an Unmet Medical Need for the Treatment of Serious Bacterial Diseases (Final), Aug-2017					√

Figure 2. Listing of documents mentioning specific randomization methodologies (* highlights mentions of variable block sizes).

Carter K, Scheffold AL, Renteria J, Berger VW, Luo YA, Chipman JJ, Sverdlov O. Regulatory guidance on randomization and the use of randomization tests in clinical trials: a systematic review. *Stat Biopharm Res*. 2024;16(4):428–440. doi:10.1080/19466315.2023.2239521

Which randomization methods are used in practice?

		2014 (n=167)	2014 incl. HTA (n=196)	2019 (n=385)
Allocation Method	Purely Random	1 (1)	2 (1)	25 (6)
	Blocked	15 (10)	20 (11)	35 (9)
	Stratified or Stratified Block	123 (79)	135 (74)	244 (64)
	Minimisation/Covariate adaptive	14 (9)	23 (13)	57 (15)
	Other	2 (1)	3 (1)	24 (6)

- The authors screened all trial documents and contacted the authors if the randomization method was unclear (therefore method only unknown in 2 trials)
- Use of **block randomization still dominant**, but goes down (85% → 73%)

Bruce CL, Juszczak E, Ogollah R, Partlett C, Montgomery A. A systematic review of randomisation method use in RCTs and association of trial design characteristics with method selection. BMC Med Res Methodol 2022;22(1):314. doi:10.1186/s12874-022-01786-4

Regulatory guidelines on randomization

- **Regulatory guidelines continue to emphasize permuted block randomization**

➡ This may implicitly suggest it is the preferred or recommended method, despite advances in randomization methodology.

Why does "randomized" alone not guarantee valid results?

Literature

History of randomization

- Silverman WA, Chalmers I. Casting and drawing lots: a time honoured way of dealing with uncertainty and ensuring fairness. BMJ. 2001 Dec 22-29;323(7327):1467-8. doi: 10.1136/bmj.323.7327.1467. PMID: 11751358; PMCID: PMC1121910.
- Fisher, R. A. (1935). *The design of experiments.* Oliver & Boyd.
- Hill, A. B. (1937). *Principles of Medical Statistics*. London: The Lancet Limited.
- Streptomycin treatment of pulmonary tuberculosis: a Medical Research Council investigation. BMJ 2:769-782. Year 1948; doi:10.1136/bmj.2.4582.769
- Doll R (2001). The role of data monitoring committees. In: Duley L, Farrell B (editors) *Clinical trials*. London: BMJ Books, p97.
- *British Medical Journal Editorial Board. The controlled therapeutic trial. Br Med J. 1948;2:791-792.*
- Matthews RA. The problematic history of randomised controlled trials Part 1: presumption and confusion on the road to randomisation. *J R Soc Med*. 2026;119(1):25–30. doi:10.1177/01410768261419044
- Matthews RA. The problematic history of randomised controlled trials Part 2: Hill's 'pragmatic' view of randomisation and its origins. *J R Soc Med*. 2026;119(2):55–58. doi:10.1177/01410768261419045
- Matthews RA. The problematic history of randomised controlled trials Part 3: Mainland, Hill and the future of RCTs. *J R Soc Med*. 2026;119(3):83–88. doi:10.1177/01410768261419062
- The James Lind Library. Illustrating the development of fair tests of treatments in health care. https://www.jameslindlibrary.org/ (accessed 2 September 2026).
- Senn S (2013). Seven myths of randomisation in clinical trials. Statistics in Medicine 32(9):1439-1450. 10.1002/sim.5713

General literature on randomization

- Rosenberger WF, Uschner D, Wang Y. Randomization: The forgotten component of the randomized clinical trial. Stat Med. 2019;38(1):1–30
- Rosenberger WF, Lachin JM. Randomization in Clinical Trials: Theory and Practice. New York: Wiley; 2002.
- Bruce CL, Juszczak E, Ogollah R, Partlett C, Montgomery A. A systematic review of randomisation method use in RCTs and association of trial design characteristics with method selection. BMC Med Res Methodol 2022;22(1):314. doi:10.1186/s12874-022-01786-4
- A Short Introduction to randomizeR, Diane Uschner, May 2026. https://dianeu.github.io/randomizeR2.0/
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Regulatory view on randomization

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- ICH E9: Statistical Principles for Clinical Trials. ICH Harmonised Tripartite Guideline, 1998.

Block II: Bias & transparent reporting

- (Allocation) Bias in randomized clinical trials
- What to report in the trial protocol and publication



Time	Session	Topics
8:30-8:40 (10 min)	Welcome & workshop overview	• Overview of the workshop topics and flow
8:40-9:40 (60 min)	I — Block I — General introduction to randomization	• History and evolution of randomization • Randomization, allocation concealment, and blinding • Randomization procedures and their key characteristics • Regulatory perspectives on randomization
9:40-10:30 (50 min)	II — Block II — Bias & transparent reporting	• (Allocation) Bias in randomized clinical trials, • Transparent reporting: what to report in the trial protocol and publication
10:30-11:00 (30 min)	Coffee Break	
11:00-12:15 (75 min)	III — Block III — Randomization in complex trial designs	• Randomization in small-sample group-sequential designs • (Stratified) randomization and allocation bias in multi-endpoint trials • Randomization in multi-arm and platform trials
12:15-12:30 (15 min)	IV — Block IV — Recommendations & take-home messages	• Key lessons learned and further topics • Open-research question and working groups

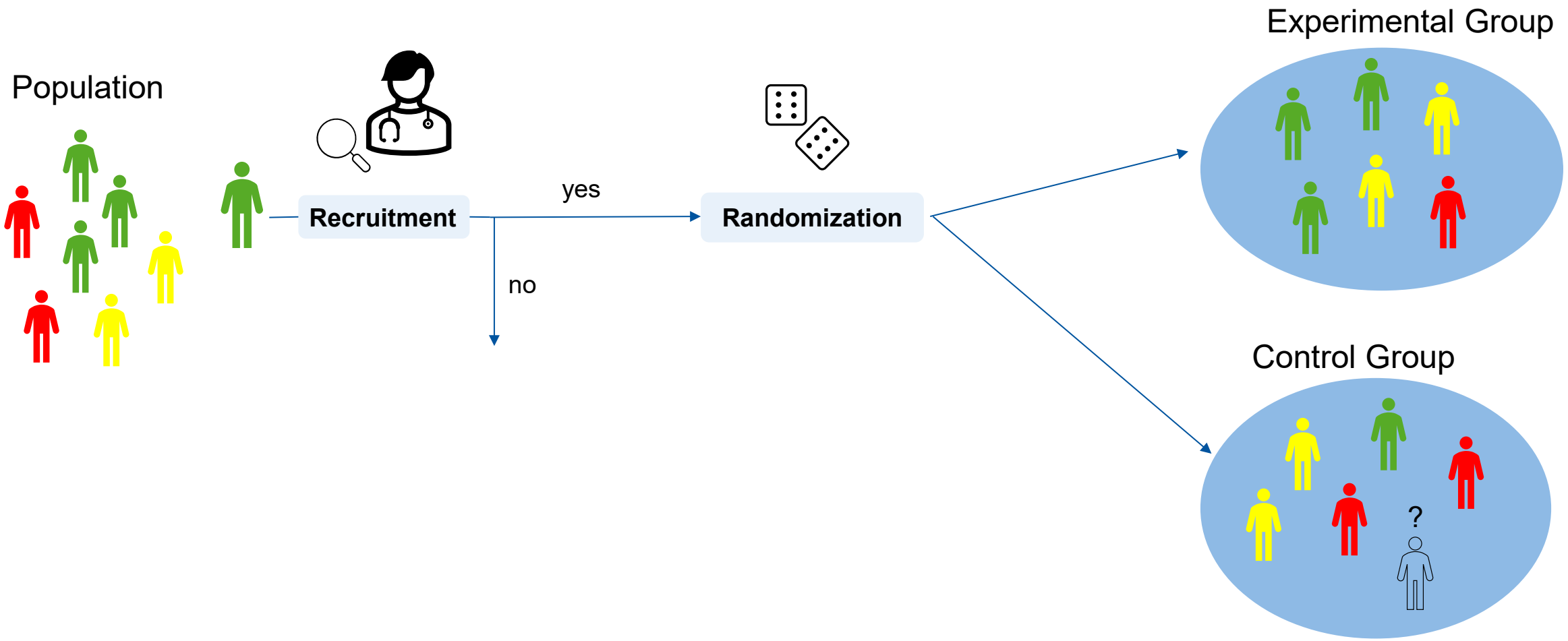
Let us talk about bias in randomized clinical trials

- **Allocation Bias:** A type of selection bias that occurs when future treatment allocations can be predicted from previous ones, usually due to improper blinding

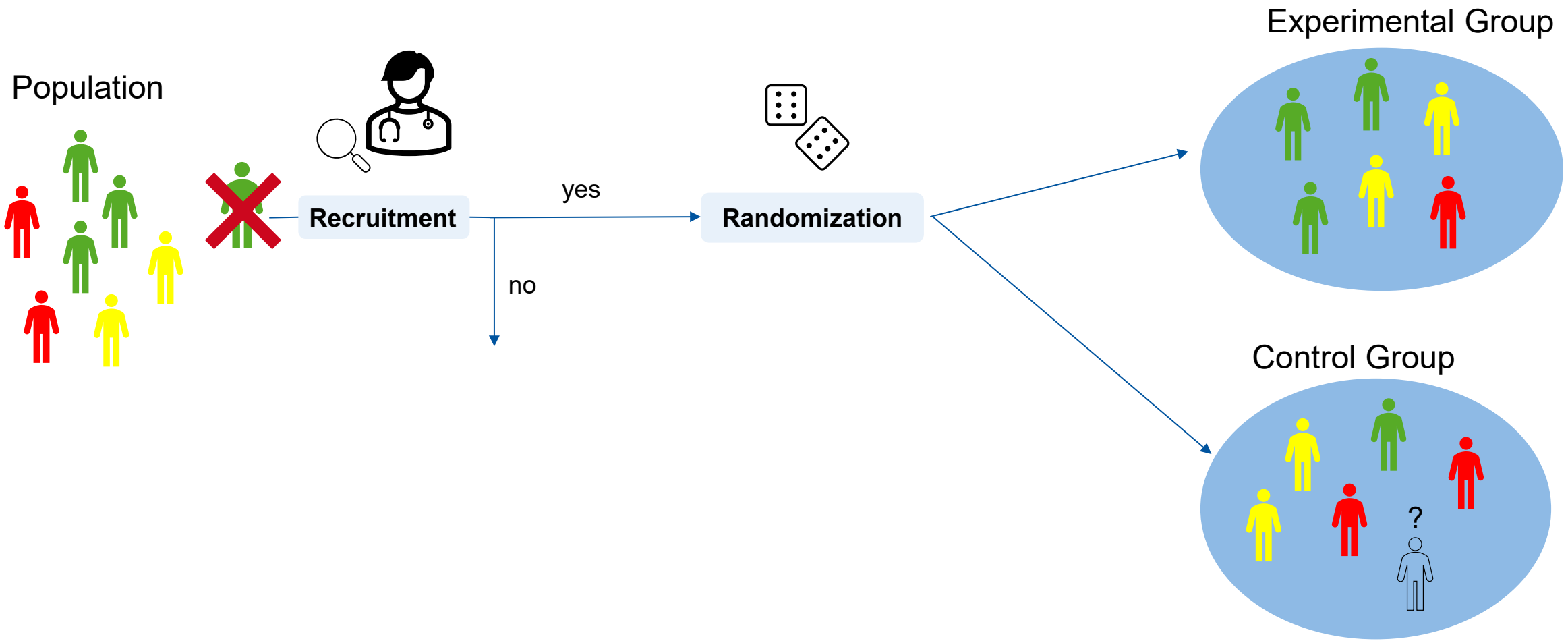
- **Chronological Bias:** Systematic differences if patients recruited earlier in the study and those recruited later

https://catalogofbias.org/

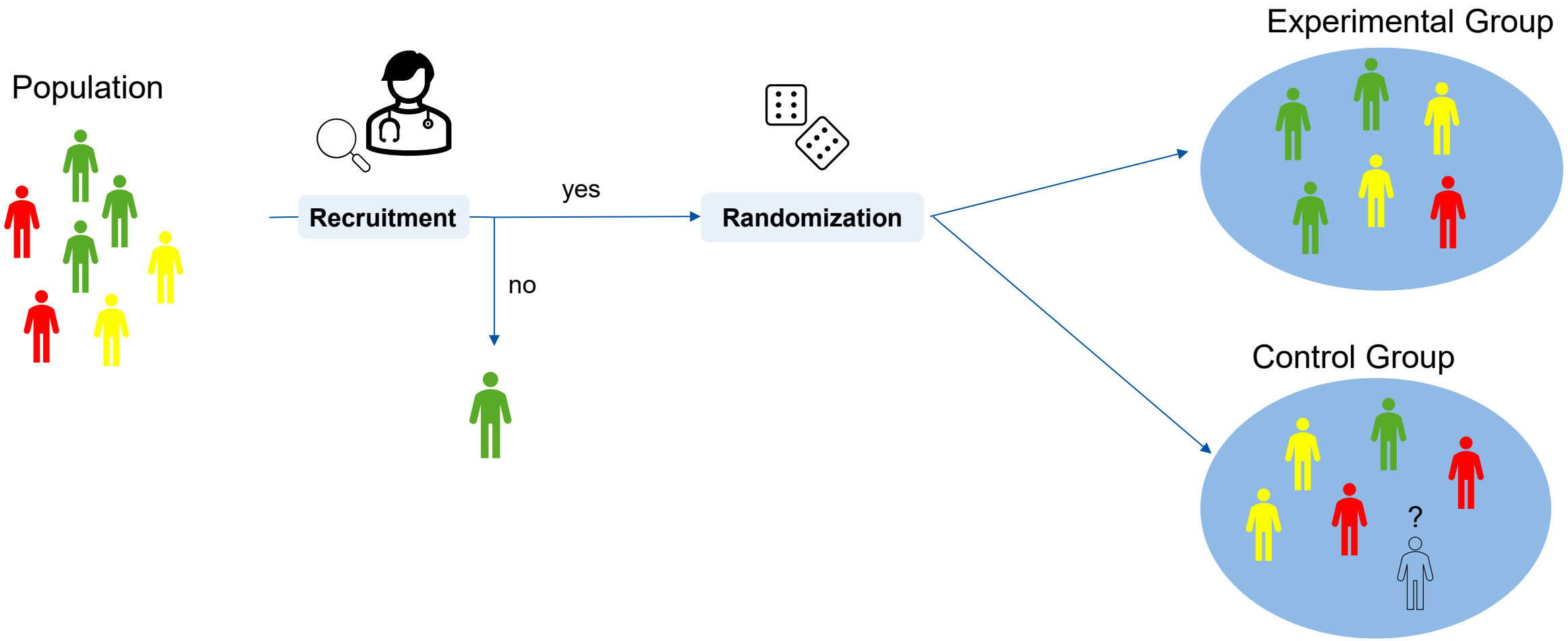
Allocation bias



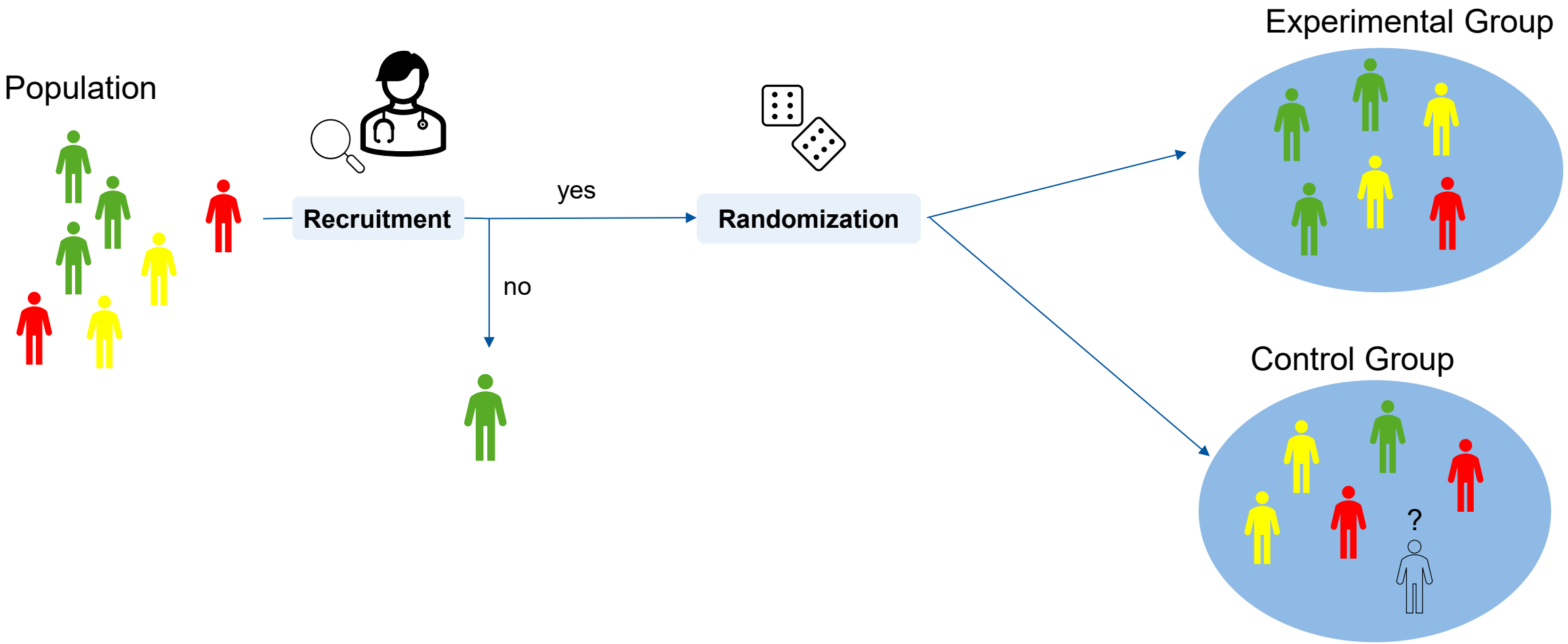
Allocation bias



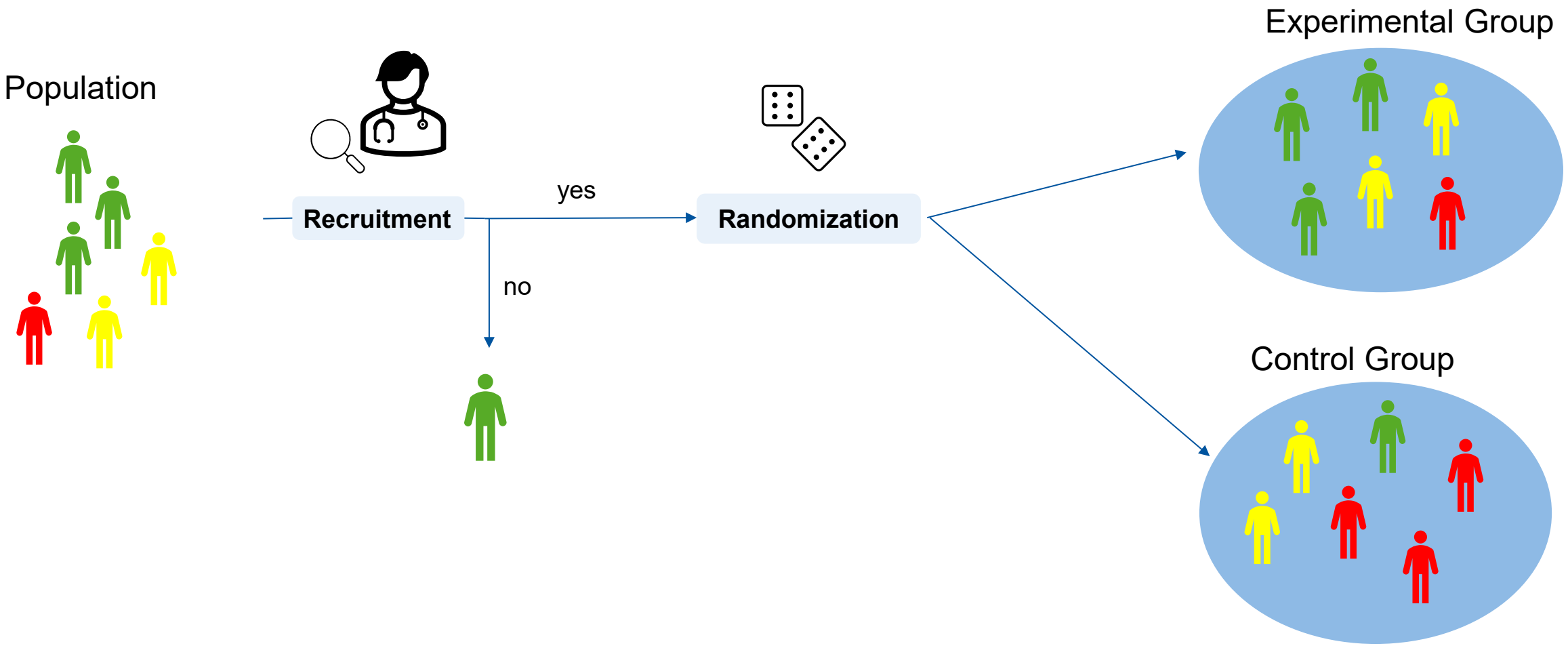
Allocation bias



Allocation bias



Allocation bias



Qualitative assessment of bias: RoB 2 tool

RoB 2 is used to assess the risk of bias in randomized trials included in Cochrane Reviews.

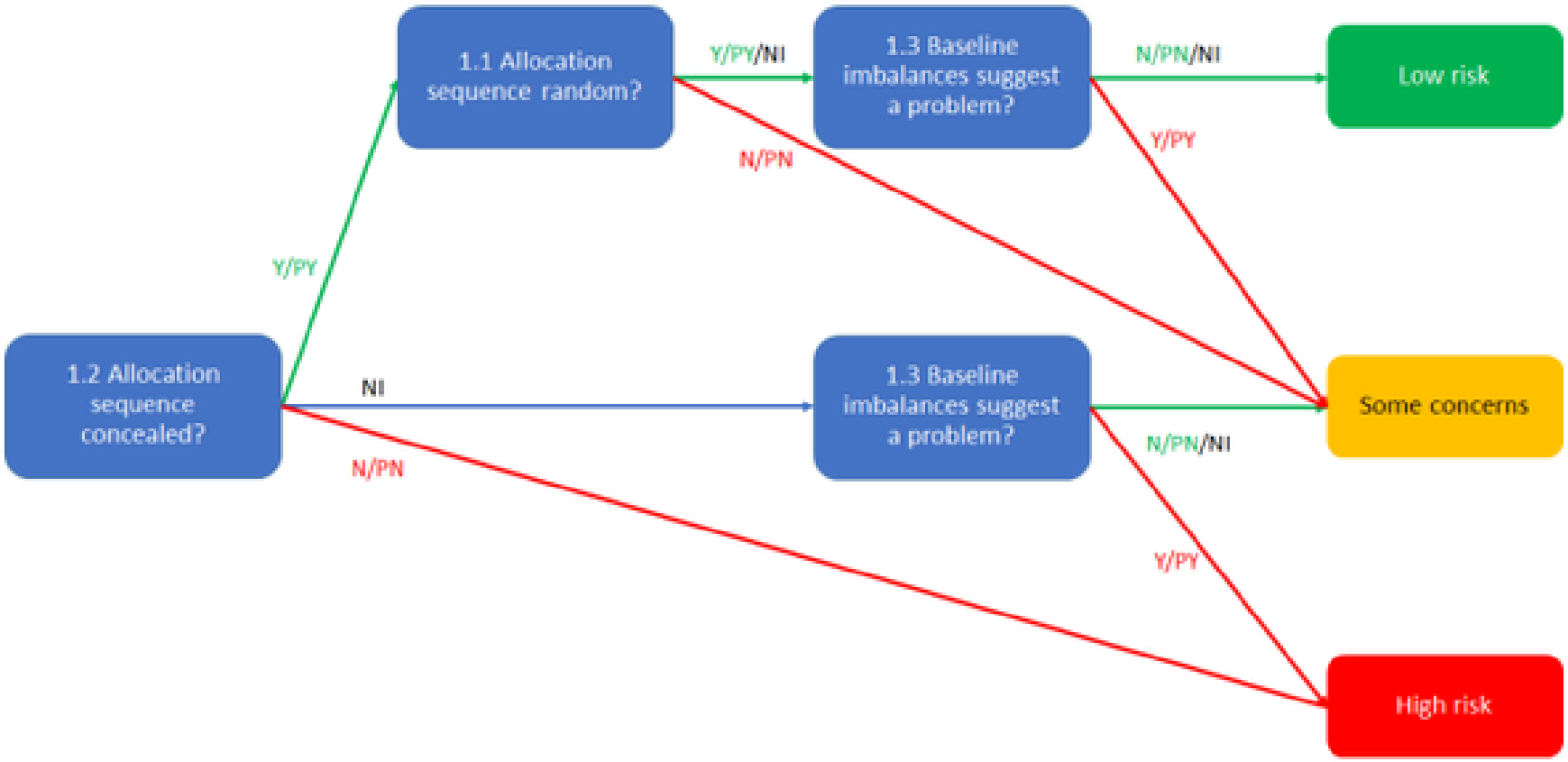
Risk of bias is assed over five domains:

1. Randomization process
2. Deviations from intended interventions
3. Missing outcome data
4. Measurement of the outcome
5. Selection of the reported result

Sterne, J. A. C., Savović, J., Page, M. J., Elbers, R. G., Blencowe, N. S., Boutron, I., et al. (2019). RoB 2: A revised tool for assessing risk of bias in randomised trials. BMJ, 366, l4898. https://doi.org/10.1136/bmj.l4898

Qualitative assessment of bias: RoB 2 tool

Figure 1. Algorithm for suggested judgement of risk of bias arising from the randomization process.



Y/PY = yes / probably yes • N/PN = no / probably no • NI = no information

Sterne, J. A. C., Savović, J., Page, M. J., Elbers, R. G., Blencowe, N. S., Boutron, I., et al. (2019). RoB 2: A revised tool for assessing risk of bias in randomised trials. BMJ, 366, l4898. https://doi.org/10.1136/bmj.l4898

Biased patient responses

Model for two-arm parallel group design, continuous endpoint

$$\boldsymbol{y_i} = \boldsymbol{\mu_E} \mathrm{t}_i + \boldsymbol{\mu_C}(\mathbf{1} - \mathrm{t}_i) + \boldsymbol{\tau_i} + \boldsymbol{\epsilon_i}$$

$$1 \leq i \leq N = N_E + N_C$$

- Two groups: Experimental (E) and Control (C)
- y_i response of patient i
- t_i allocation for each patient i with $\mathrm{t}_i = 1$ if patient i is allocated to E and $\mathrm{t}_i = 0$ otherwise
- τ_i allocation bias effect
- $\epsilon_i \sim N(0, \sigma^2)$ error term

Allocation bias policy

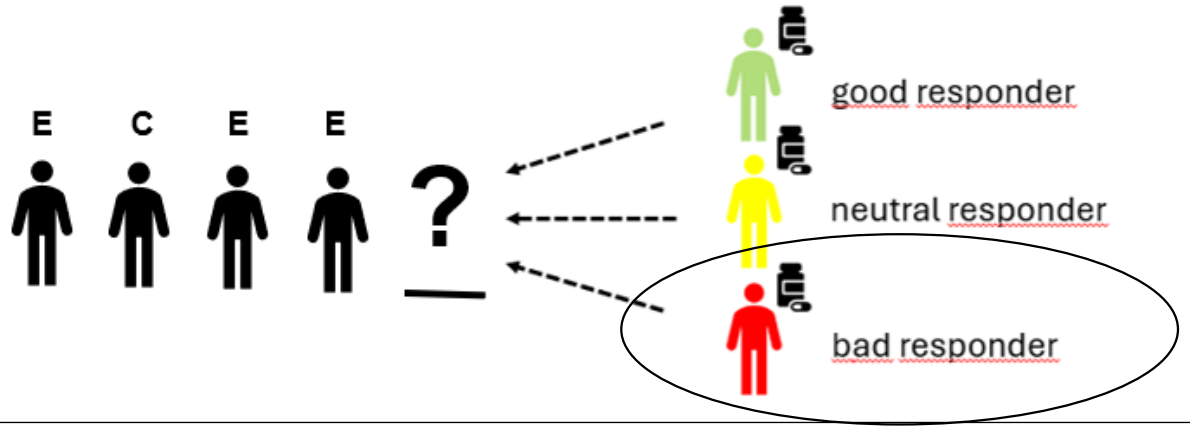
Settings:

- Researcher knows previous allocations and predicts future assignments
- Favors the experimental group E
- Better-responding patients are associated with higher expected outcomes.
- Guessing follows **the convergence strategy of Blackwell & Hodges**[1]

Allocation bias policy

$$\tau_j = \begin{cases} \eta, & if \quad \mathrm{n}_E(j-1) < \mathrm{n}_C(j-1) \\ 0, & if \quad \mathrm{n}_E(j-1) = \mathrm{n}_C(j-1) \\ -\eta, & if \quad \mathrm{n}_E(j-1) > \mathrm{n}_C(j-1) \end{cases}$$

- η : allocation bias effect
- $\mathrm{n}_i(j)$: Number of patients in group $\mathrm{i} \in \{E, C\}$ after j assignments



[1] Blackwell, D., & Hodges, J. L., Jr. (1957). Design for the control of selection bias. The Annals of Mathematical Statistics, 28(2), 449–460. https://doi.org/10.1214/aoms/1177706973

Why does allocation bias matter?

Simulation setting

- two-arm parallel group design
- continuous normal endpoint
- n=32

- null hypothesis: $\mu_E = \mu_C$
- statistical test: t-test
- 80% power
- 5% significance level
- allocation bias effect: 10% of the treatment effect associated with 80% power
- 16 000 randomization lists

Randomization Procedure [RP]	Type-I Error [mean]	$P_{RP}(w \leq 0.05)$
CR	0.050	0.464
RAR	0.051	0.139
PBR(4)	0.056	0.000
BSD(4)	0.050	0.372
BSD(2)	0.052	0.086
EBC(2/3)	0.052	0.108
CHEN(4,2/3)	0.052	0.087

Allocation bias distort inference in randomized trials, increase the type-I-error rates and decrease the validity of trial results !

Wied S, Razsolkov DK, Heussen N, Hilgers R-D (2026) Does randomization matter in rare disease clinical trials. PLoS One 21(9): e0339427. https://doi.org/10.1371/journal.pone.0339427

Why does bias matter?

> *"The interpretation of statistical measures of uncertainty of the treatment effect and treatment comparisons should involve consideration of the potential contribution of bias to the p-value, confidence interval, or inference."*
>
> ICH E9

The impact of bias on inference should be evaluated, and it should be chosen a trial design that mitigate allocation bias best!

ICH E9: Statistical Principles for Clinical Trials. ICH Harmonised Tripartite Guideline, 1998.

Evaluated study designs

	Parallel group design	Endpoint	Hypothesis	Test	Allocation Bias	Chronological Bias
Hilgers et al. [8] (2017)	2-arm single center	continuous	$H_0 : \mu_E = \mu_C$	t-Test	additive	stepwise, linear, logarithmic
Hilgers et al. [9] (2019)	2-arm multicenter	continuous	$H_0 : \mu_E = \mu_C$	stratified t-Test	additive	stepwise, linear, logarithmic
Uschner et al. [10] (2018)	multi-arm single center	continuous	$H_0 : \mu = 0$	ANOVA	additive	-
Rückbeil et al. [11] (2017)	2-arm single center	time to event	$H_0 : \lambda_E/\lambda_C = 1$ $(H_0 : S_E = S_C)$	F-/Logrank-Test	multiplicative	stepwise, linear, logarithmic
Schoenen et al. [39] (2024)	2-arm single center	continuous	$H_{0,k} : \mu_{E,k} = \mu_{C,k}$ $(k \in \{1, \ldots, m\},\ m \geq 2)$	t-Test (Šidák, All-or-None)	additive	-
Schoenen et al. [40] (2026)	2-arm multicenter	continuous	$H_0 : \mu_E = \mu_C$ $(\mu_E, \mu_C \in \mathbb{R}^m,\ m \geq 2)$	Wei-Lachin Test (stratified)	additive	-
Bodden et al. [37] (accepted)	2-arm group sequential	continuous	$H_0 : \mu_E = \mu_C$	t-Test	additive	-
Bodden et al. [38] (under Review)	multi-arm platform	continuous	$H_0 : \mu_E = \mu_C$	t-Test	additive	stepwise, linear, inverted-U, seasonal
Ihl et al. (work ongoing)	2-arm multicenter	binary	$H_0 : \pi_E = \pi_C$	Fisher-Test Boshloo-Test	multiplicative	-

Table 1. Overview of already evaluated study designs with respect to randomization procedures and bias.

Wied S, Razsolkov DK, Heussen N, Hilgers R-D (2026) Does randomization matter in rare disease clinical trials. PLoS One 21(9): e0339427. https://doi.org/10.1371/journal.pone.0339427

Chronological bias

Chronological bias occurs when the treatment effect is confounded with an **unobserved time trend** over the enrollment sequence

Reason is that over long recruitment periods the following may change:

- referral patterns
- Recruitment policy policy
- diagnostic standards
- staff experience
- background treatments

> *"Although unrestricted randomisation is an acceptable approach, some advantages can generally be gained by randomising subjects in blocks. This helps to increase the comparability of the treatment groups, particularly when subject characteristics may change over time, as a result, for example, of changes in recruitment policy."*
>
> ICH E9, Section 2.3.2

Chronological bias

Tamm & Hilgers (2014) evaluated **chronological bias for different block sizes**

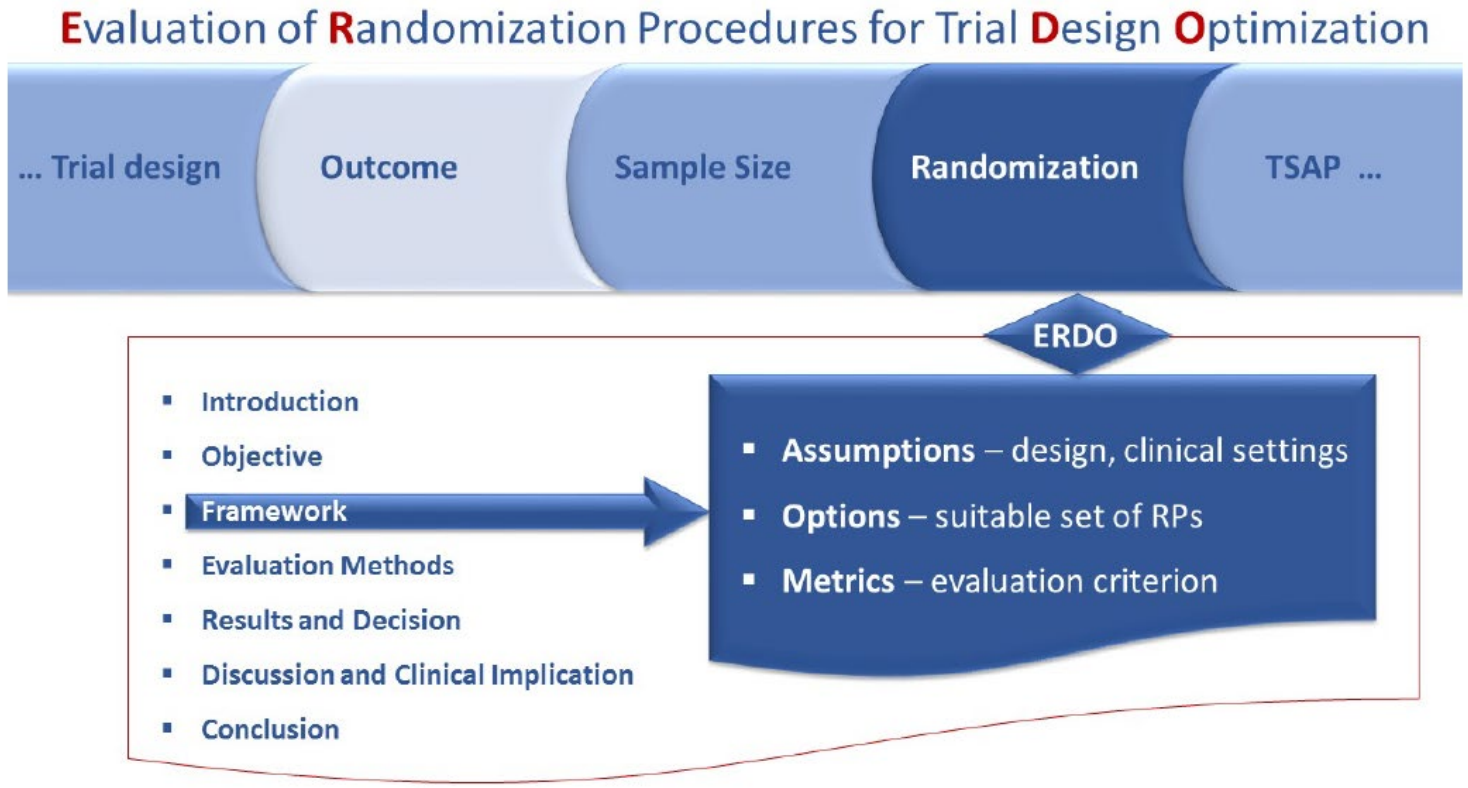
- over all sequences, the t-test becomes conservative
- conditional on an unfavorable realized sequence, type I error can be heavily inflated
- smaller blocks restrict the inflated type I error for unfavorable realized sequences
- smaller blocks lead to mean T1E becoming more conservative

Tamm M, Hilgers R-D (2014). Chronological bias in randomized clinical trials arising from different types of unobserved time trends. *Methods of Information in Medicine* 53(6):501–510.

Regulatory agencies recognize that whenever the allocation ratio varies over calendar time a time trend in the outcome becomes confounded with treatment, which is explicitly stated for response-adaptive randomization [ICH E20 §4.5] and for non-concurrent controls and arms entering or leaving a platform [FDA, 2023].

ICH (2025). *E20: Adaptive Designs for Clinical Trials*, Step 2b draft, §4.5.
FDA (2023, revised draft June 2026). *Master Protocols for Drug and Biological Product Development: Guidance for Industry*.

ERDO template



Benchmark process of the choice of the RP

Hilgers RD, Uschner D, Rosenberger WF, Heussen N. ERDO - a framework to select an appropriate randomization procedure for clinical trials. BMC Med Res Methodol. 2017 Dec 4;17(1):159. doi: 10.1186/s12874-017-0428-z. PMID: 29202708; PMCID: PMC5715815.

Transparent reporting of randomization

STUDY PROTOCOL Open Access

Corticosteroids versus clobazam in epileptic encephalopathy with ESES: a European multicentre randomised controlled clinical trial (RESCUE ESES*)

Bart van den Munckhof[1], Alexis Arzimanoglou[2,3], Emilio Perucca[4], Heleen C. van Teeseling[5], Frans S. S. Leijten[6], Kees P. J. Braun[1], Floor E. Jansen[1*] and on behalf of the RESCUE ESES study group

Treatment

Patients will be enrolled by their treating physician/study doctor, who has no direct insight in the allocation mechanism of the randomisation module. Treatment is allocated according to an automatic online randomisation module with block randomisation stratified for centre to the two treatment regimens (1:1 ratio). The randomisation module was programmed by a data manager. The patients and treating physicians are not blinded for treatment allocation, as this was considered not feasible with two treatment arms that differ in their prescription form (in most centres, corticosteroids are given as intravenous monthly pulses, whilst clobazam is given as a tablet to be taken daily) and because the two treatments require different monitoring. The neuropsychological assessment and EEG assessment will be performed by personnel blinded for treatment:

van den Munckhof B et al. Corticosteroids versus clobazam in epileptic encephalopathy with ESES: a European multicentre randomised controlled clinical trial (RESCUE ESES*). Trials. 2020 Nov 23;21(1):957. doi: 10.1186/s13063-020-04874-2. PMID: 33228736; PMCID: PMC7686710.

Corticosteroids versus clobazam for treatment of children with epileptic encephalopathy with spike-wave activation in sleep (RESCUE ESES): a multicentre randomised controlled trial

Marleen M L van Arnhem, Bart van den Munckhof*, Alexis Arzimanoglou, Emilio Perucca, Liisa Metsähonkala, Guido Rubboli, Marianne Søndergaard Khinchi, Anne de Saint-Martin, Kerstin A Klotz, Julia Jacobs, J Helen Cross, Irene Garcia Morales, Wim M Otte, Heleen C van Teeseling, Frans S S Leijten, Kees P J Braun, Floor E Jansen, on behalf of the RESCUE ESES study group†*

Randomisation and masking

A data manager, independent of the study group, generated an online randomisation scheme with random permuted blocks (block sizes of one and two), and this scheme was uploaded into a secure online database (Research Online version 2) to ensure concealment of treatment allocation. Block randomisation was stratified for centre to ensure equal allocation to each treatment group. Participants were randomly assigned (1:1) to treatment with corticosteroids or clobazam. Participants and treating physicians were not masked to the treatment assignment, because of the inherent differences in mode of administration and expected adverse effects between treatment groups. The neuropsychologists and neurophysiologist (EEG readers) who evaluated the outcome were masked to treatment assignment.

van Arnhem MML et al. Corticosteroids versus clobazam for treatment of children with epileptic encephalopathy with spike-wave activation in sleep (RESCUE ESES): a multicentre randomised controlled trial. Lancet Neurol. 2024 Feb;23(2):147-156. doi: 10.1016/S1474-4422(23)00409-X. Epub 2023 Dec 8. PMID: 38081201.

Which randomization methods are used in practice?

Kahan & Morris (2012): 258 RCTs published in BMJ, JAMA, Lancet and NEJM (2010)

Table 2 Reporting and analysis of trials

Variables	No (%) of trials (n=258)
Method of randomisation:	
Simple randomisation	4 (2)
Permuted blocks without stratification	40 (16)
Permuted blocks with stratification	85 (33)
Minimisation	29 (11)
Other	4 (2)
Unclear	96 (37)

- **The method was unclear in 37% of trials**

Kahan BC, Morris TP. Reporting and analysis of trials using stratified randomisation in leading medical journals: review and reanalysis. BMJ 2012;345:e5840. doi:10.1136/bmj.e5840

Bias when reporting of random sequence generation and allocation concealment is unclear

Table 2. Impact of potential risk of bias elements on effect estimates in randomized trials

Comparison	Number of ME studies (between- or within-trial comparisons)	Average bias (95% CI)	Direction of bias (certainty of evidence[a])	Overall inference for the item
Random sequence generation				**Overestimation (Moderate certainty)**
High vs. low risk of bias	7 (between-trial)	ROR 0.93 (0.75–1.16)	Overestimation (Very low[b])	
High/unclear vs. low risk of bias	15 (between-trial)	ROR 0.94 (0.90–0.97)	Overestimation (Moderate)	
High vs. low/unclear risk of bias	2 (between-trial)	ROR 0.91 (0.76–1.09)	Overestimation (Very low[b])	
Unclear vs. low risk of bias	2 (between-trial)	ROR 0.89 (0.83–0.97)	Overestimation (Moderate)	
Allocation concealment				**Overestimation (Moderate certainty)**
High vs. low risk of bias	9 (between-trial)	ROR 0.95 (0.69–1.29)	Little or no bias (Very low[c])	
High/unclear vs. low risk of bias	16 (between-trial)	ROR 0.92 (0.88–0.97)	Overestimation (Moderate)	
High vs. low/unclear risk of bias	2 (between-trial)	ROR 0.92 (0.80–1.07)	Overestimation (Very low[b])	
Unclear vs. low risk of bias	3 (between-trial)	ROR 0.85 (0.74–0.98)	Overestimation (Moderate)	
Definitely/probably unconcealed vs. definitely/probably concealed	1 (between-trial)	ROR 0.87 (0.66–1.14)	Overestimation (Very low[b,d])	

Where concealment or sequence generation was only unclear, effects were still overestimated: ROR 0.85 (0.74–0.98) and 0.89 (0.83–0.97), both moderate certainty.

Wang Y, Parpia S, Couban R, et al. Compelling evidence from meta-epidemiological studies demonstrates overestimation of effects in randomized trials that fail to optimize randomization and blind patients and outcome assessors. J Clin Epidemiol. 2024;165:111211. doi:10.1016/j.jclinepi.2023.11.001

Transparent reporting of randomization

Often trial reports only stated that patients were *"randomized"*

- Some trials gave vague descriptions, such as:
  - "Computer-generated random numbers were used"
  - "A random numbers table was used"
- Important details were often missing:
  - Who generated the random sequence
  - How allocation was concealed
  - Which randomization method was used
- Poor reporting makes it difficult to evaluate the risk of bias

Clear reporting of randomization is essential for assessing trial quality.

Kahan B C, Morris T P. Reporting and analysis of trials using stratified randomisation in leading medical journals: review and reanalysis *BMJ* 2012; 345 :e5840 doi:10.1136/bmj.e5840

SPIRIT 2025

What information regarding randomization should be provided in the study protocol?

Methods: Assignment of interventions		
Randomization:		
Sequence generation	21a	Who will generate the random allocation sequence and the method used
	21b	Type of randomization (simple or restricted) and details of any factors for stratification. To reduce predictability of a random sequence, other details of any planned restriction (e.g., blocking) should be provided in a separate document that is unavailable to those who enroll participants or assign interventions
Allocation concealment mechanism	22	Mechanism used to implement the random allocation sequence (e.g., central computer/telephone; sequentially numbered, opaque, sealed containers), describing any steps to conceal the sequence until interventions are assigned
Implementation	23	Whether the personnel who will enroll and those who will assign participants to the interventions will have access to the random allocation sequence
Blinding	24a	Who will be blinded after assignment to interventions (e.g., participants, care providers, outcome assessors, data analysts)
	24b	If blinded, how blinding will be achieved and description of the similarity of interventions
	24c	If blinded, circumstances under which unblinding is permissible, and procedure for revealing a participant's allocated intervention during the trial

The protocol should be transparent, but it should not compromise allocation concealment.

For example, avoid giving recruiter access to:

- The full randomization list
- Random seed
- Parameter of the randomization procedures, i.e. block length

Chan AW, Boutron I, Hopewell S, et al. SPIRIT 2025 statement: updated guideline for protocols of randomised trials. BMJ 2025;389:e081477. doi:10.1136/bmj-2024-081477

What to Report in the Trial Publication?

Randomization:		
Sequence generation	17a	Who generated the random allocation sequence and the method used
	17b	Type of randomization and details of any restriction (eg, stratification, blocking and block size)
Allocation concealment mechanism	18	Mechanism used to implement the random allocation sequence (eg, central computer/telephone; sequentially numbered, opaque, sealed containers), describing any steps to conceal the sequence until interventions were assigned
Implementation	19	Whether the personnel who enrolled and those who assigned participants to the interventions had access to the random allocation sequence
Blinding	20a	Who was blinded after assignment to interventions (eg, participants, care providers, outcome assessors, data analysts)
	20b	If blinded, how blinding was achieved and description of the similarity of interventions

Do not only write: "Participants were randomized."

CONSORT expects enough detail to judge whether allocation was unbiased, concealed, correctly implemented, and appropriately analyzed.

Hopewell S, Chan AW, Collins GS, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. BMJ 2025;389:e081123. doi:10.1136/bmj-2024-081123

Literature

Bias in RCTs

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Allocation bias

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- Blackwell D, Hodges JL. Design for the Control of Selection Bias. Ann Math Stat. 1957;28(2):449–60.
- Uschner D, Schindler D, Hilgers RD, Heussen N. randomizeR: An R Package for the Assessment and Implementation of Randomization Procedures. J Stat Softw 2018;85(8):1–22.
- Wied S, Razsolkov DK, Heussen N, Hilgers R-D (2026) Does randomization matter in rare disease clinical trials. PLoS One 21(9): e0339427. https://doi.org/10.1371/journal.pone.0339427
- Proschan, M. (1994). INFLUENCE OF SELECTION BIAS ON TYPE I ERROR RATE UNDER RANDOM PERMUTED BLOCK DESIGNS. *Statistica Sinica*, *4*(1), 219–231. http://www.jstor.org/stable/24305282

Chronological Bias

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- Tamm M, Hilgers R-D (2014). Chronological bias in randomized clinical trials arising from different types of unobserved time trends. *Methods of Information in Medicine* 53(6):501–510. doi:10.3414/me14-01-0048
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- Hilgers R-D, Uschner D, Rosenberger WF, Heussen N (2017). ERDO — a framework to select an appropriate randomization procedure for clinical trials. *BMC Medical Research Methodology* 17:159.
- ICH (1998). *E9: Statistical Principles for Clinical Trials*, §2.3.2.
- ICH (2025). *E20: Adaptive Designs for Clinical Trials*, Step 2b draft, §4.5.
- FDA (2023, revised draft June 2026). *Master Protocols for Drug and Biological Product Development: Guidance for Industry*.
- Bofill Roig M, Krotka P, Burman C-F, et al. (2022). On model-based time trend adjustments in platform trials with non-concurrent controls. *BMC Medical Research Methodology* 22:228.

Reporting of randomization in clinical trials

- Hopewell S, Chan AW, Collins GS, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. BMJ 2025;389:e081123. doi:10.1136/bmj-2024-081123
- Chan AW, Boutron I, Hopewell S, et al. SPIRIT 2025 statement: updated guideline for protocols of randomised trials. BMJ 2025;389:e081477. doi:10.1136/bmj-2024-081477
- ICH E9: Statistical Principles for Clinical Trials. ICH Harmonised Tripartite Guideline, 1998.
- Kahan BC, Morris TP. Reporting and analysis of trials using stratified randomisation in leading medical journals: review and reanalysis. BMJ 2012;345:e5840. doi:10.1136/bmj.e5840

10:30-11:00

COFFEE BREAK



Block III: Randomization in Complex Trial Designs

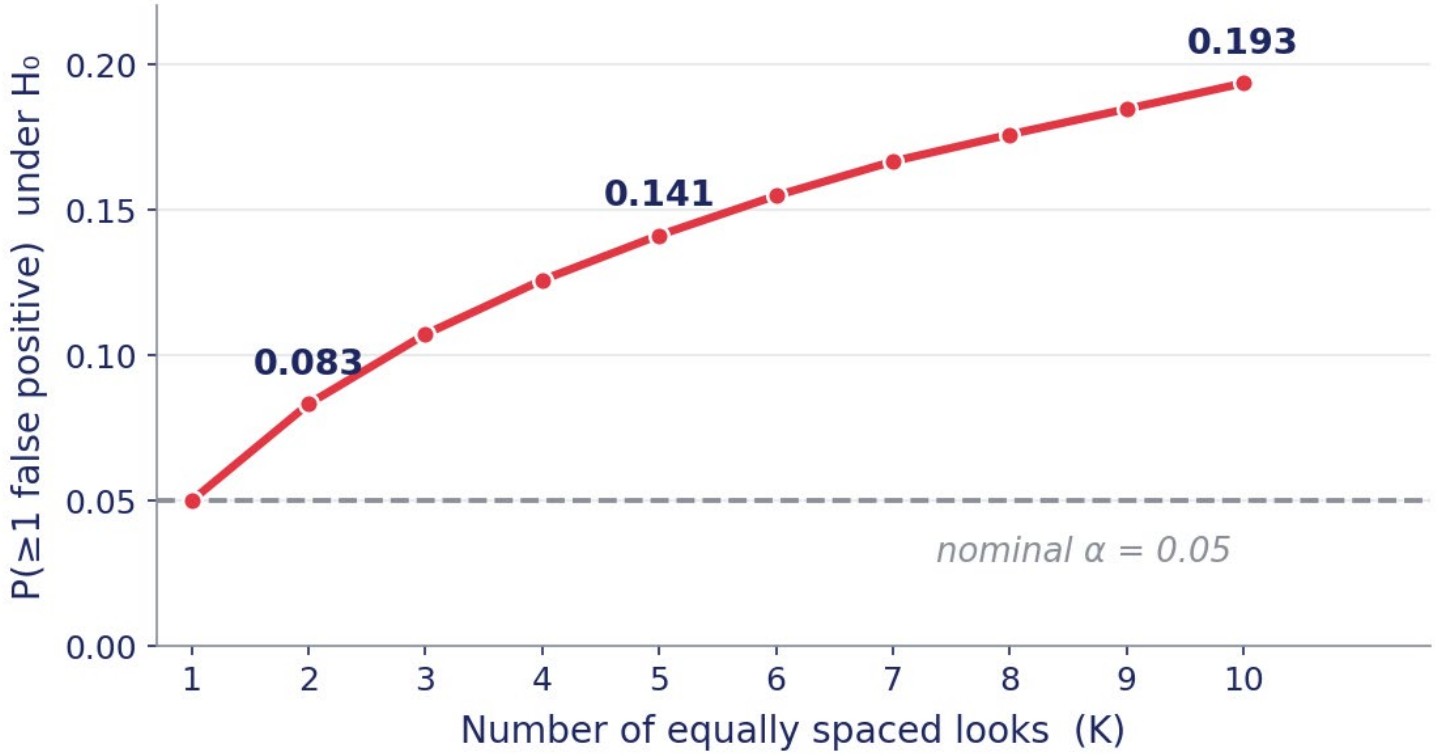
- Randomization in small-sample group-sequential designs
- (Stratified) randomization and allocation bias in multi-endpoint trials
- Randomization in multi-arm and platform trials

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12:15-12:30 (15 min)	IV Block IV — Recommendations & take-home messages	• Key lessons learned and further topics • Open-research question and working groups

Interim analysis

„An interim analysis is any analysis intended to compare treatment arms with respect to efficacy or safety at any time prior to formal completion of a trial. “ (ICH E9)

- Interim analyses introduce a multiple testing problem
- Adjustment for multiplicity is therefore required



Reproduces Armitage, McPherson & Rowe (1969), J. R. Statist. Soc. A 132:235–244.

Definitions for group sequential designs

Group sequential design

- Randomized clinical trial with pre-planned interim analyses
- Allows early trial termination for efficacy and/or futility

(Efficacy) boundaries

Pocock
- Equal significance level spent for each stage

O'Brien-Fleming
- Low significance level in early stages
- Gradual increase with each test

Types of group sequential designs

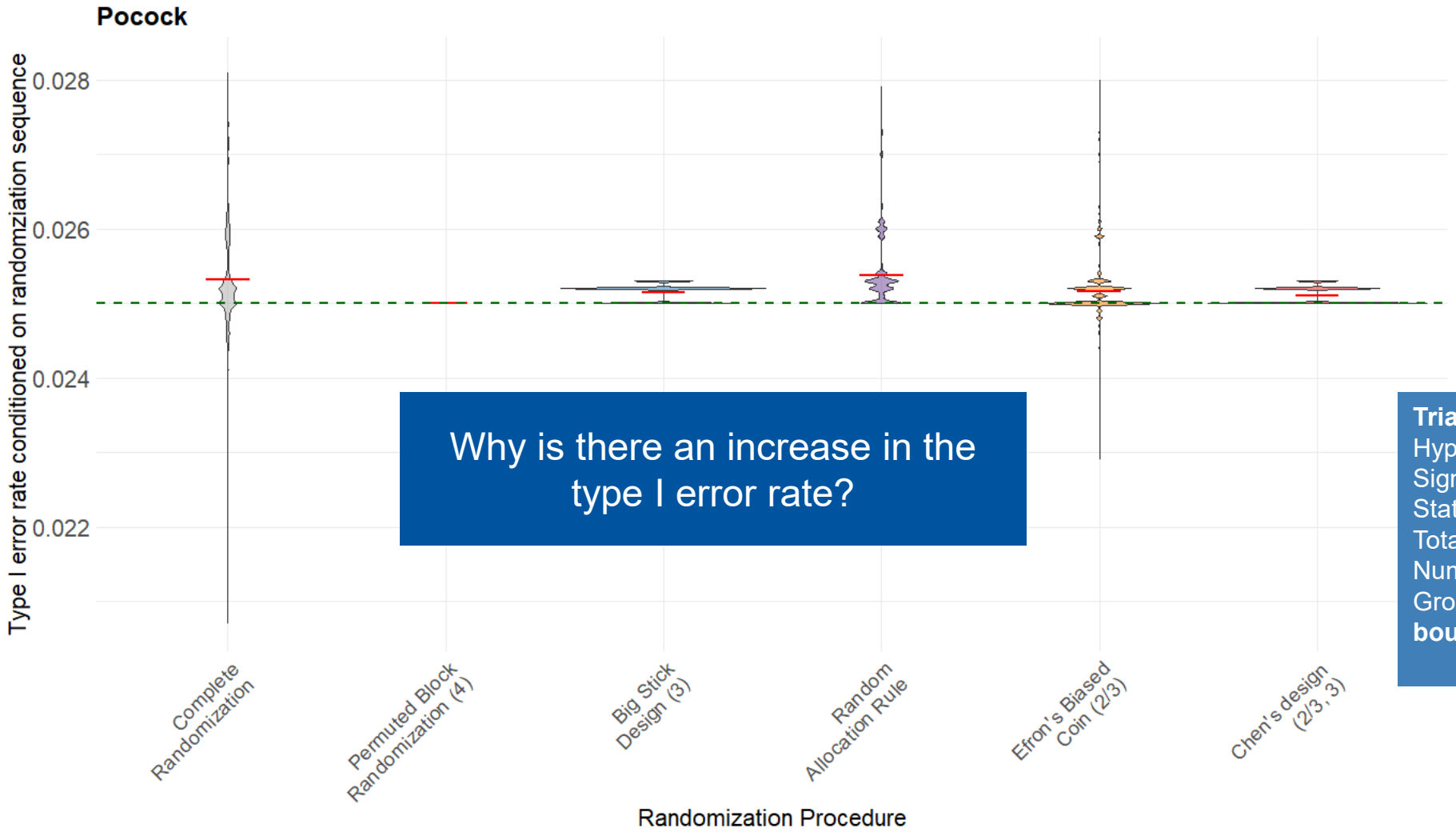
Lan-DeMets
- Uses a **pre-defined alpha spending function**
- Allows for
  - adjustments in the timing and number of analyses
  - adjustments for unequal stage-wise sample sizes
- Alpha spending function can mimick **Pocock** or **O'Brien-Fleming** type boundaries

Inverse Normal Combination Test
- Combines stage-wise test statistics
- Uses a predefined combination function
- Allows
  - data-dependent adaptions
  - usage of standard group sequential boundaries

Randomization in group sequential designs

How do different randomization procedures influence the validity of group sequential trials in small sample clinical trials?

Type I error rate conditioned on randomization sequence

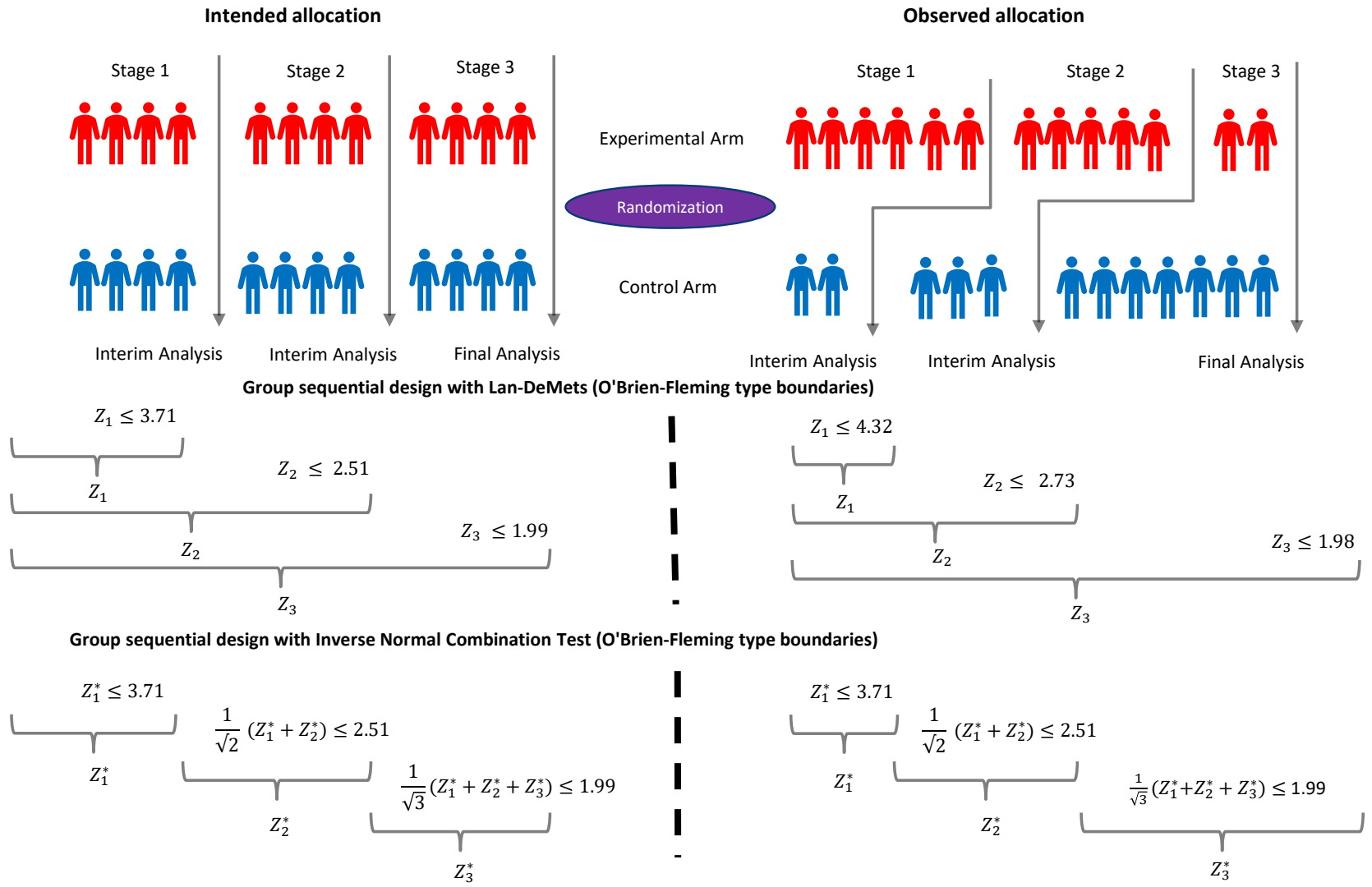


Why is there an increase in the type I error rate?

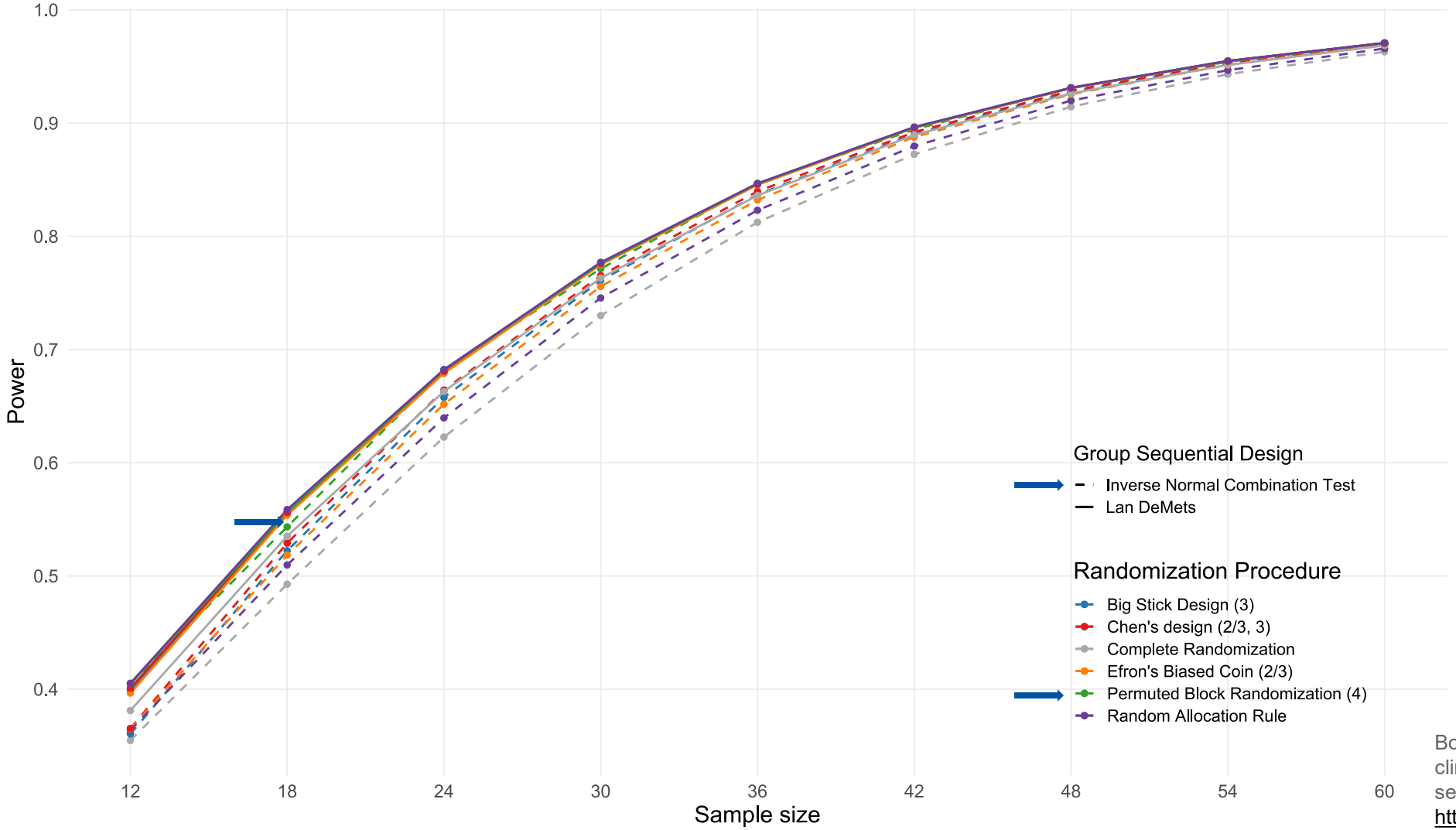
Trial setup
Hypothesis: $H_0: \mu_1 \leq \mu_2$
Significance level: $\alpha = 0.025$
Statistical test: **z-test**
Total number of patients: **24**
Number of interim analyses: **2**
Group sequential design: **Standard Pocock boundaries with allocation ratio of 1:1**

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Observed allocation vs intended allocation



Power of different randomization procedures by sample size



Hypothesis: $H_0: \mu_1 \leq \mu_2$
Significance level: $\alpha = 0.025$
Statistical test: **z-test**
Number of interim analyses: **2**
Type of boundaries: **O'Brien-Fleming type** (Lan-DeMets alpha spending function)
Standardized effect size: **1.0**
Power calculation Interval: **Every 6 patients**

Bodden D, Hilgers R-D, König F. Randomization in clinical trials with small sample sizes using group sequential designs. PLOS ONE, 13 June 2025. https://doi.org/10.1371/journal.pone.0325333

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Framework for selecting the randomization procedure for (very) small stage-wise sample sizes

Randomization procedure / group sequential design	Standard boundaries*	Lan-DeMets using observed information	Inverse Normal Combination Test
Complete Randomization	Type I error not controlled	Low power; risk of allocations to single group	Low power; risk of allocations to single group
Random Allocation Rule	Type I error not controlled	Risk of allocations to single group	Low power; risk of allocations to single group
Big Stick Design with maximum tolerated imbalance b	Type I error not controlled	For $b \leq \frac{n}{K}$	For $b \leq \frac{2n}{K}$ power reduction relative to 1:1 allocation per stage
Permuted Block Randomization with block size a	a must divide $\frac{n}{K}$ to control type I error		a should divide $\frac{n}{K}$ for optimal power
Efron‘s Biased Coin	Type I error not controlled	Risk of allocations to single group	Moderate power; risk of allocations to single group
Chen‘s Design with maximum tolerated imbalance b, probability p	Type I error not controlled	For $b \leq \frac{n}{K}$	For $b \leq \frac{2n}{K}$ power reduction relative to 1:1 allocation per stage

*With standard boundaries we refer to the case where the boundaries are calculated assuming a 1 : 1 allocation ratio
Legend: n = total sample size, K = number of stages, b = maximum tolerated imbalance, p = probability for biased coin

Which randomization methods are used in group sequential trials?

Bodden, Hilgers & König (2025): randomization procedure in 71 group sequential trials (Ovid MEDLINE, 2019–2023)

Randomization Procedure	n = 71
(Randomized) Permuted Block Randomization	30 (42)
No information on block length	16
Information on block length(s)	14
4	6
10	2
20	1
3, 6	1
4, 6	1
2, 4, 6	1
2, 4, 6, 8	1
4, 6, 8, 10	1
Minimization	9 (13)
Simple Randomization	1 (1)
No information given	31 (44)

- **Permuted blocks used in 42% of trials**
- **No information on the randomization procedure in 44% of trials**

Bodden D, Hilgers R-D, König F. Randomization in clinical trials with small sample sizes using group sequential designs. PLOS ONE, 13 June 2025. https://doi.org/10.1371/journal.pone.0325333

How large is the impact of allocation bias in group sequential trials?

- Permuted block randomization can keep an exact 1:1 allocation ratio at every interim analysis
- It is also most commonly used in group sequential trials
- But we know it is suspectable to allocation bias

So how high is that allocation bias risk actually?

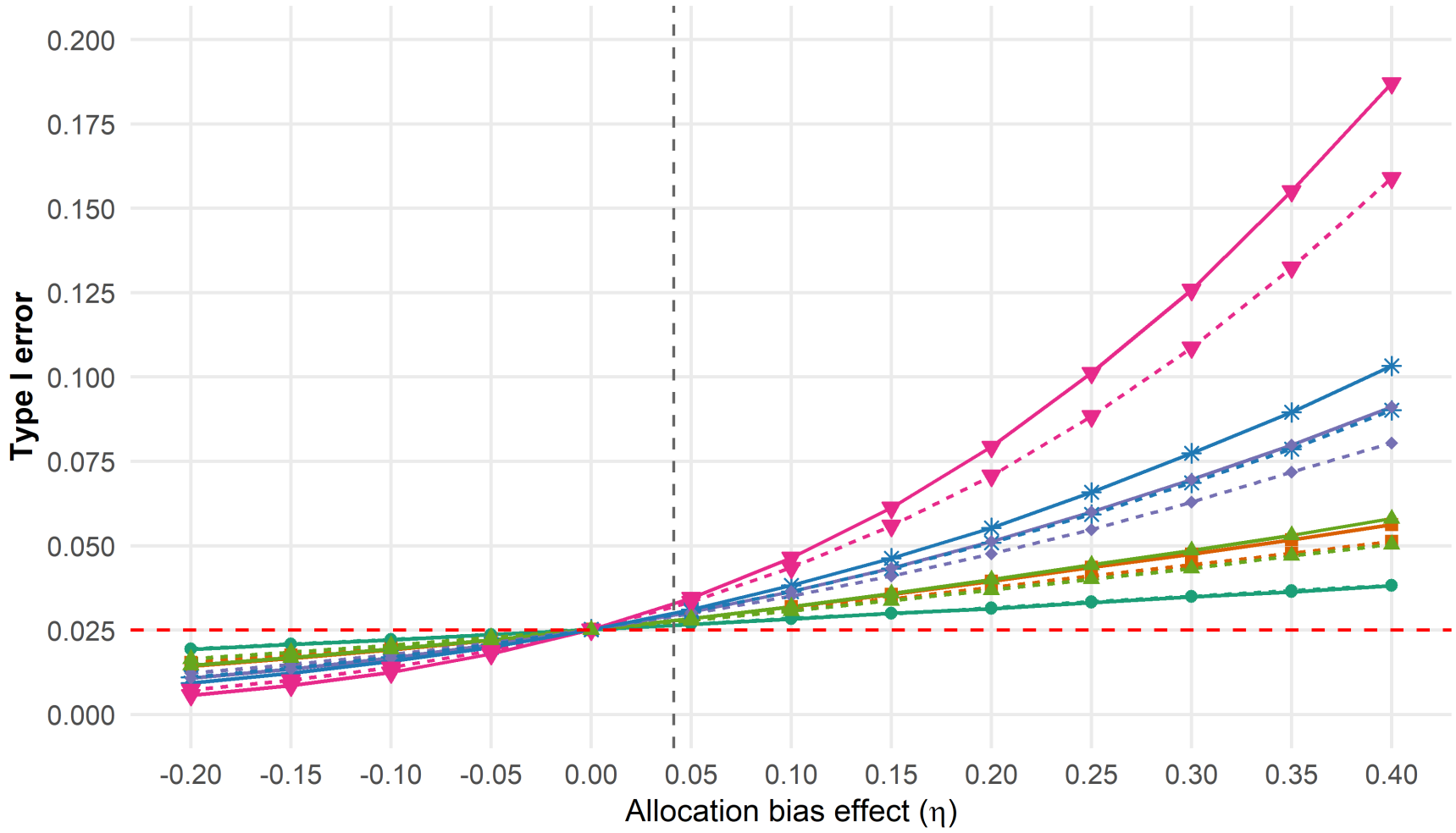
Allocation bias in group sequential trials

Simulation Setup

- **Design:** Two-arm parallel group trial
- **Endpoint:** Normal (continuous)
- **Sample size:** 48 patients
- **Statistical test:** One-sided t-test, interim analysis after 24 patients
- **Simulation:** 16 000 randomization sequences × 1 000 trials per sequence

Bodden D, Hilgers RD, König F. When randomization is not random: Allocation bias in small sample, group sequential randomized clinical trials. Stat Methods Med Res. 2026 Jun;35(6):1295-1309. doi: 10.1177/09622802261442914. Epub 2026 Apr 29. PMID: 42057417.

One efficacy interim analysis



Observations

- **Permuted Block Randomization (4)** shows largest type I error inflation
- Pocock-type spending functions lower bias impact than O'Brien-Fleming-type
- **Big Stick Design (3)** controls type I error fairly well
- Negative bias (investigator favors control) leads to conservative type I error rates

Bodden D, Hilgers RD, König F. When randomization is not random: Allocation bias in small sample, group sequential randomized clinical trials. Stat Methods Med Res. 2026 Jun;35(6):1295-1309. doi: 10.1177/09622802261442914. Epub 2026 Apr 29. PMID: 42057417.

Literature

- Bodden, Hilgers & König (2025). Randomization in clinical trials with small sample sizes using group sequential designs. PLOS ONE.
- Bodden, Hilgers & König (2026). When randomization is not random: allocation bias in small sample, group sequential trials. Stat Methods Med Res.
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(Stratified) randomization and allocation bias in multiple endpoint trials

Motivation

- ICH E9 recommends using a **single primary endpoint.**
- **Rare diseases**: Heterogeneous, multifaceted diseases, small populations

  → one primary endpoint often insufficient
- **Multiple endpoints:** More comprehensive; may increase statistical power.

Objective

Is an increasing number of endpoints associated with stronger allocation bias?

Analyzing the impact of allocation bias on inference in multiple endpoint trials and investigating whether it can be minimized through an appropriate study design.

Types of multiple endpoints

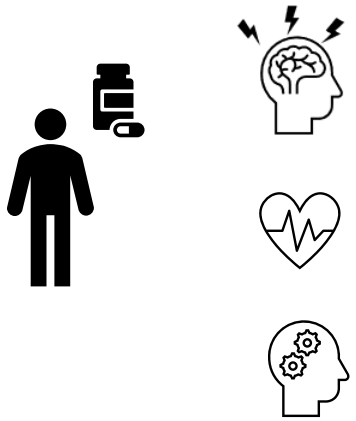
Multiple primary endpoints

Endpoints are analyzed separately, and treatment efficacy is demonstrated if at least one endpoint shows a statistically significant treatment effect

Types of multiple endpoints

Multiple primary endpoints

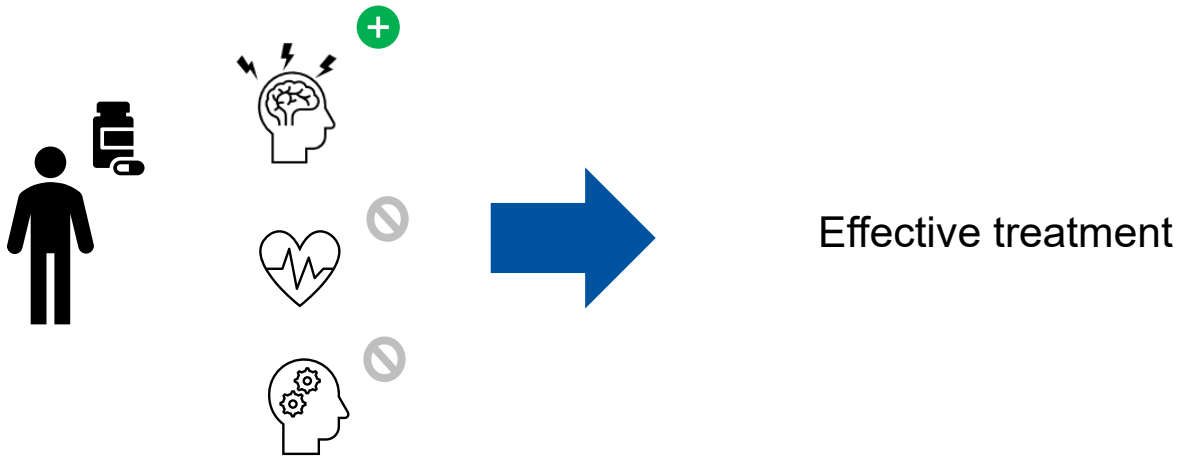
Endpoints are analyzed separately, and treatment efficacy is demonstrated if at least one endpoint shows a statistically significant treatment effect



Types of multiple endpoints

Multiple primary endpoints

Endpoints are analyzed separately, and treatment efficacy is demonstrated if at least one endpoint shows a statistically significant treatment effect

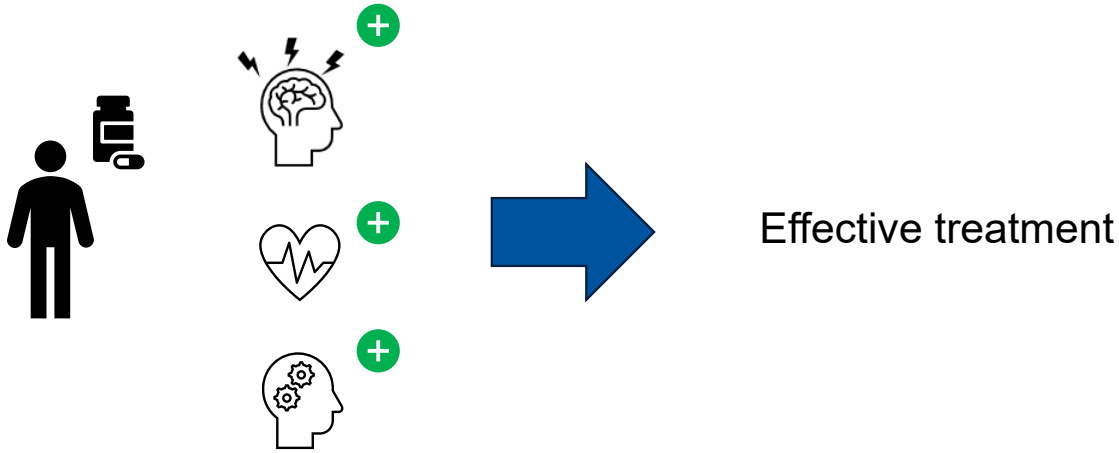


Effective treatment

Types of multiple endpoints

Co-primary Endpoints

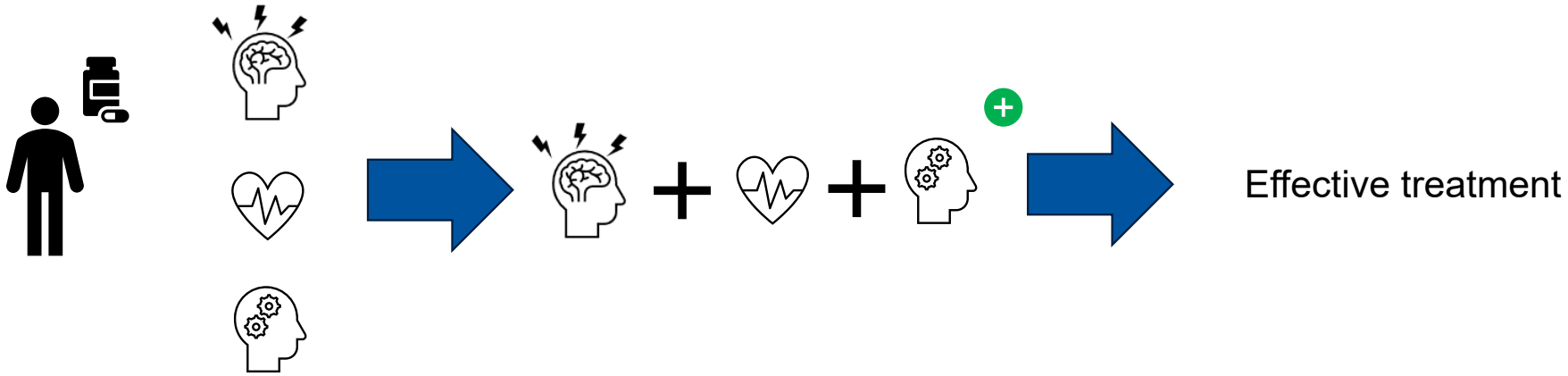
Significant treatment effects required across all endpoints to demonstrate efficacy.



Types of multiple endpoints

Multi-component Endpoint

Aggregate numerous endpoint components into a single score or rating.



Statistical model

Model for two-arm parallel group design with multiple continuous endpoints

$$\begin{pmatrix} X_{j,1} \\ X_{j,2} \\ \vdots \\ X_{j,m} \end{pmatrix} = \begin{pmatrix} \mu_{E,1} \\ \mu_{E,2} \\ \vdots \\ \mu_{E,m} \end{pmatrix} t_j + \begin{pmatrix} \mu_{C,1} \\ \mu_{C,2} \\ \vdots \\ \mu_{C,m} \end{pmatrix} (1 - t_j) + \begin{pmatrix} \tau_{j,1} \\ \tau_{j,2} \\ \vdots \\ \tau_{j,m} \end{pmatrix} + \begin{pmatrix} \varepsilon_{j,1} \\ \varepsilon_{j,2} \\ \vdots \\ \epsilon_{j,m} \end{pmatrix}$$

$$1 \leq j \leq N = n_E + n_C, m \geq 2$$

Notation:

- $X_j = (X_{j,1}, \ldots, X_{j,m})^T$: Response vector of patient j across the m endpoints.
- $t = (t_1, \ldots, t_N)^T$: Randomization list with $t_j = 1$ if patient j is allocated to E, otherwise $t_j = 0$
- $\mu_i = (\mu_{i,1}, \ldots, \mu_{i,m})^T$: Expected response vector under i ∈ {E, C}
- $\tau_j = (\tau_{j,1}, \ldots, \tau_{j,m})^T$: Allocation bias effect on responses of patient j
- $\epsilon_j \sim \mathcal{N}_m(0, \Sigma)$: Error term

Statistical inference under allocation bias

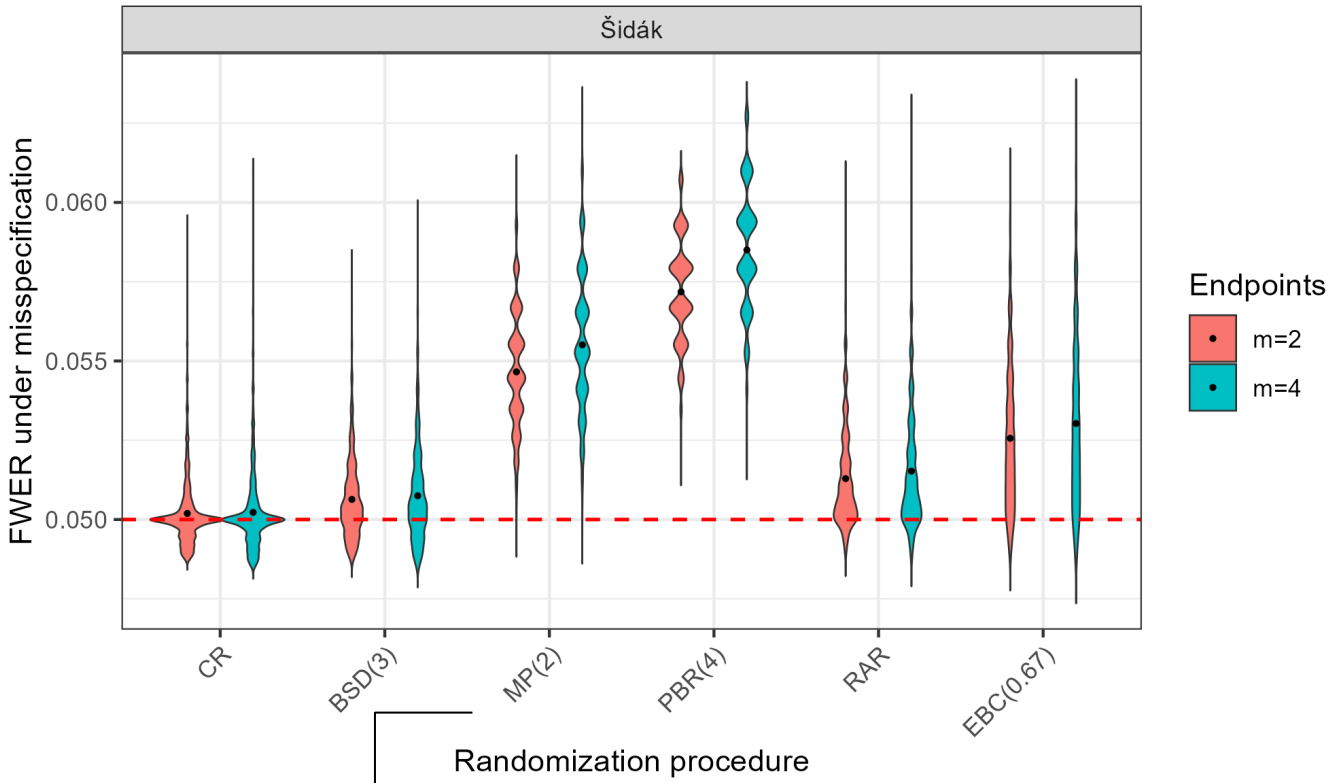
Multiple primary endpoints	Co-primary endpoints	Multi-component endpoints
At least one endpoint must show an effect • Treatment success if ≥1 endpoint is significant • Each endpoint tested separately with a two-sided t-test: $H_{0,k}: \mu_{C,k} = \mu_{E,k}$ vs. $H_{1,k}: \mu_{C,k} \neq \mu_{E,k}$ • Multiplicity adjustment required: **Šidák-adjusted significance levels**	**All endpoints must show an effect** • Treatment success only if all endpoints are significant • Each endpoint tested separately with a one-sided t-test $H_{0,k}: \mu_{C,k} = \mu_{E,k}$ vs. $H_{1,k}: \mu_{Ck} < \mu_{E,k}$ • No multiplicity adjustment necessary, **all-or-none criterion**	**Endpoint components are evaluated jointly** • Multiple endpoint components are considered jointly. • A single multivariate/global test evaluates the components: $H_0: \mu_C = \mu_E$ vs. $H_1: \mu_C < \mu_E$ • **Wei–Lachin test**

Impact of allocation bias is evaluated via T1E/FWER under model misspecification.

Evaluation of allocation bias – multiple primary endpoints

Settings:

- **Multiple primary endpoints:**
  - $m = 2,4$ endpoints
  - standard normally distributed and uncorrelated
  - evaluated using Šidák-adjusted significance levels
- **Sample size**: n= 32
- **Allocation bias effect** ~10% of the effect size associated with 80% power
- 10 000 randomization lists

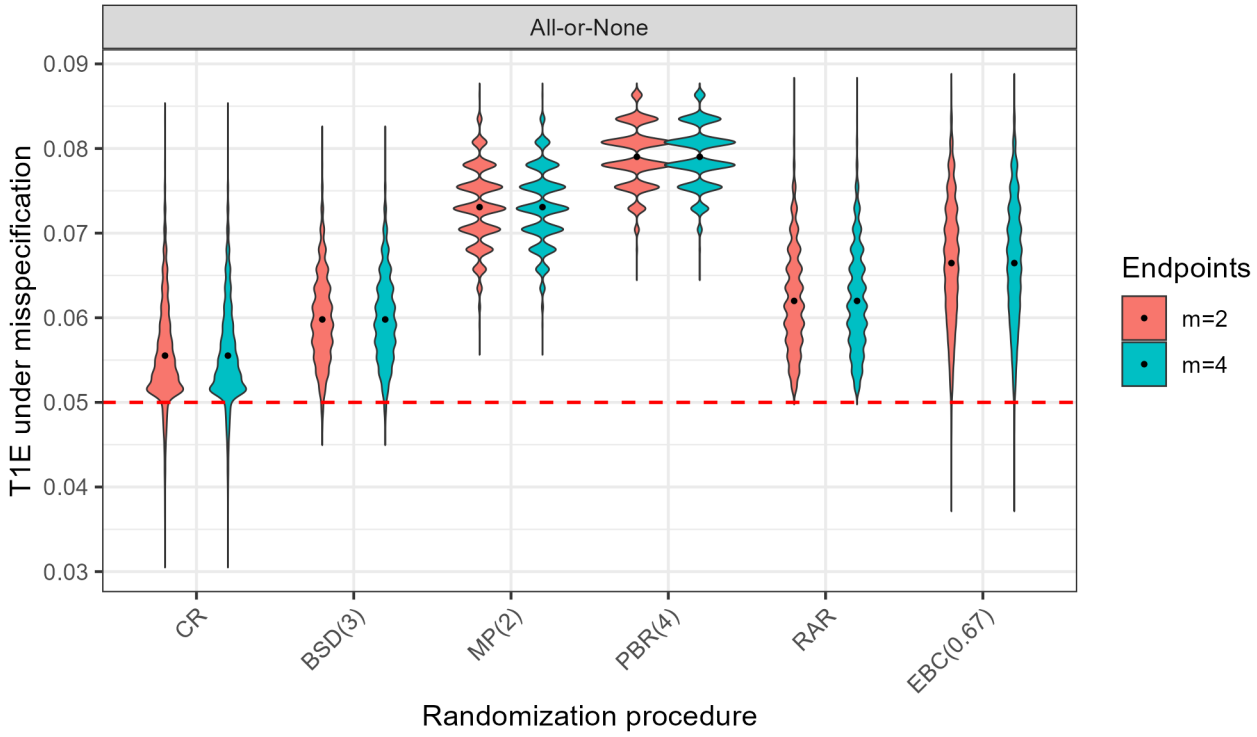


MP(b)	Maximal Procedure	Selection from randomization lists generated by CR that never exceed imbalance b.

Evaluation of allocation bias – co-primary endpoints

Settings:

- **Co-primary endpoints:**
  - $m = 2,4$ endpoints
  - standard normally distributed and uncorrelated
  - no multiplicity adjustment necessary, all-or-none criterion
- **Sample size**: n= 32
- **Allocation bias effect** ~10% of the effect size associated with 80% power
- 10 000 randomization lists

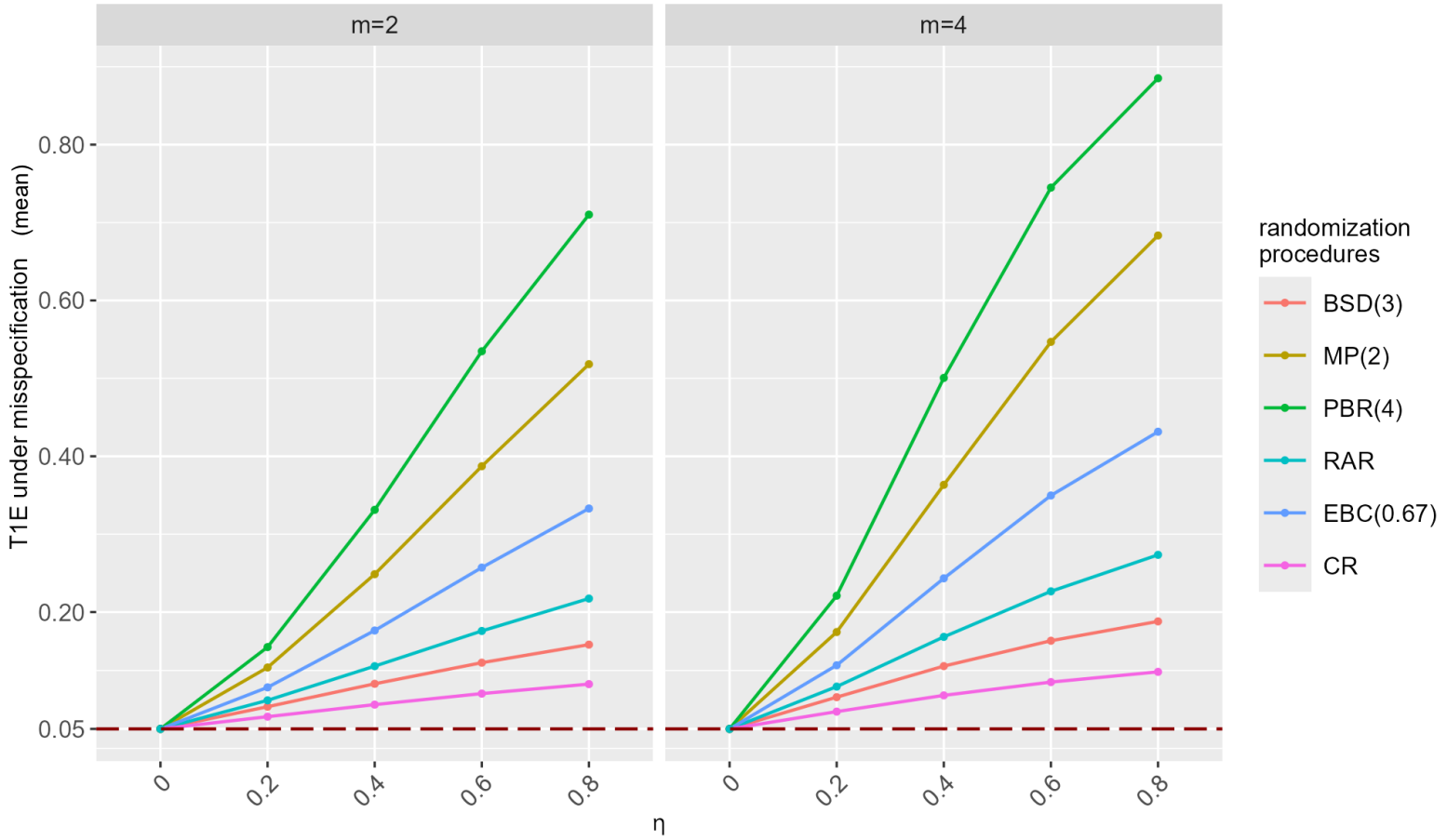


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Evaluation of allocation bias – multi-component endpoints

Settings:

- **Multi-component endpoints:**
  - $m = 2,4$ endpoints
  - standard normally distributed and uncorrelated
  - evaluated using the Wei-Lachin test
- **Sample size**: n= 32
- **Allocation bias effect** $\eta_k \in \{0,0.2,0.4,0.6,0.8\}$ for all $k \in \{1, \dots, m\}$
- 10 000 randomization lists



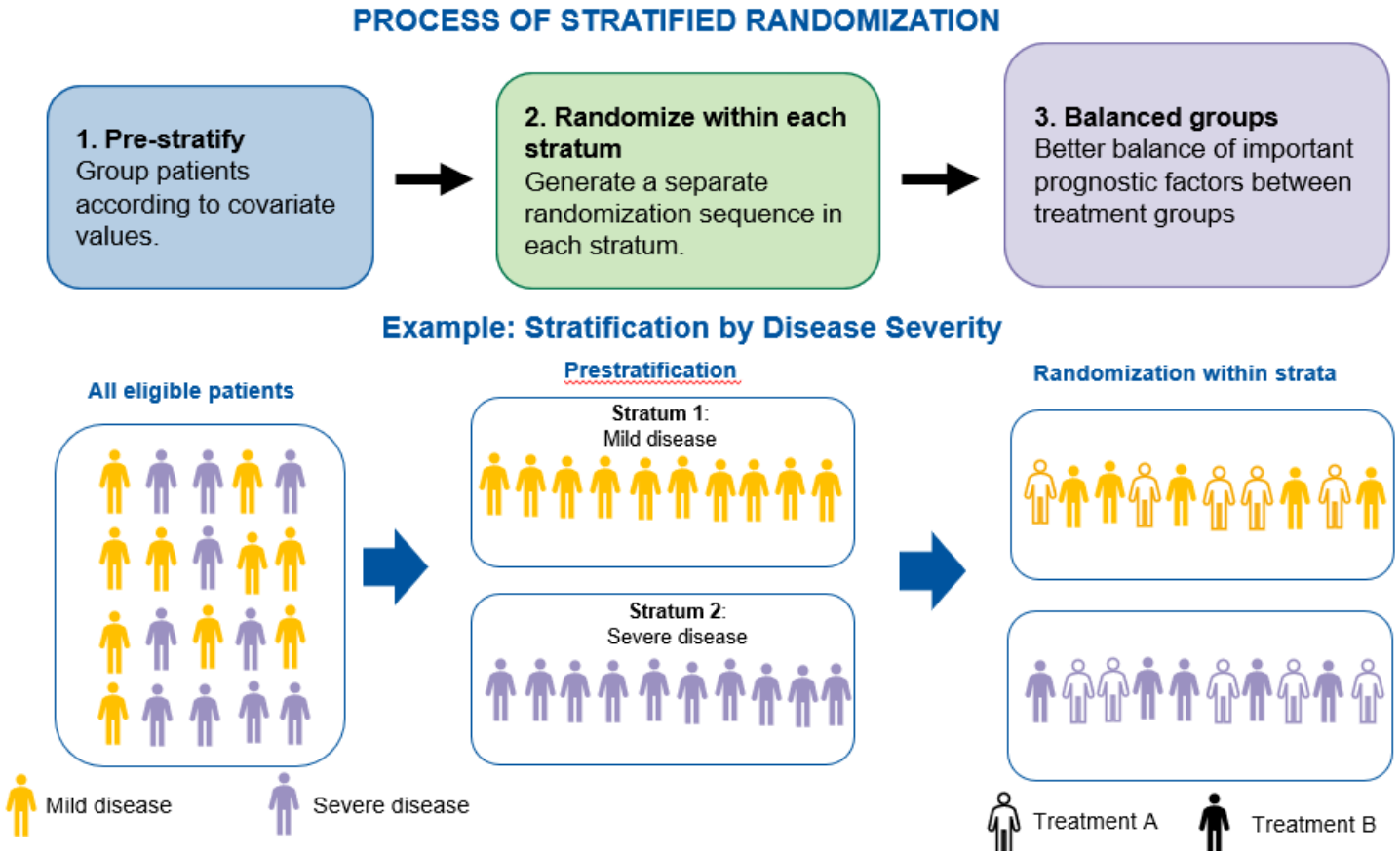
Discussion – allocation bias in trials with multiple endpoints

Allocation bias can distort inference in randomized trials with multiple endpoints.

- **The impact of allocation bias depends on several interacting factors:**
  - Endpoint structure & analysis strategy
  - Randomization procedure
  - Number of endpoint components of the multi-component endpoint
  - Correlation between endpoints

Stratified randomization (pre-stratification)

Balance within important covariate groups (strata) before randomization



Regulatory view on stratification

ICH E9 – Statistical Principles for Clinical Trials

- Stratification by **important baseline prognostic factors** (e.g. severity of disease, age, sex, etc.) can help achieve **balanced treatment allocation within strata.**
- Stratification may be particularly beneficial in **small trials.**
- The number of stratification factors should be **kept limited**; more than two or three factors is rarely necessary.
- Factors on which randomization has been stratified should be **accounted for later in the analysis**.

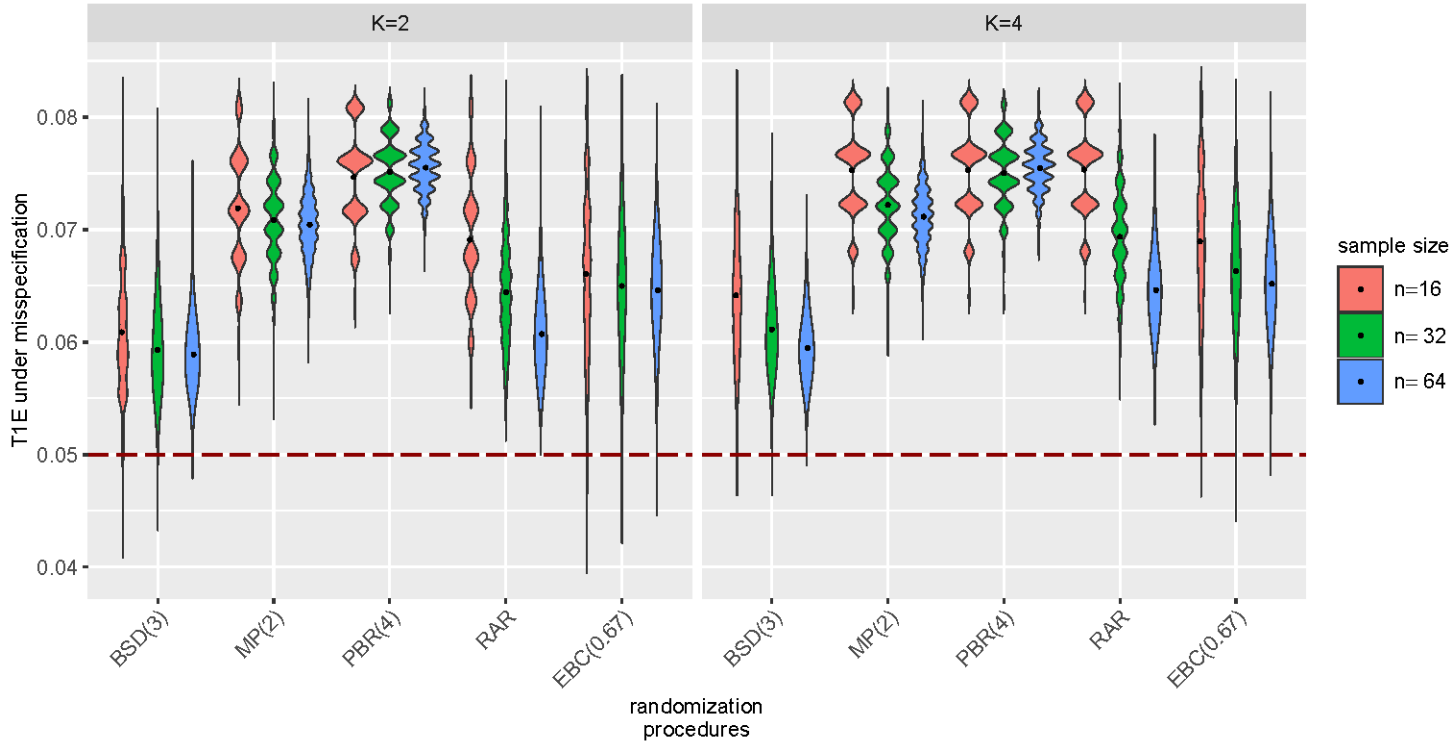
EMA Small Sample Guidance

- Stratification **improves power**, particularly if it is based on important prognostic variables.
- Factors used to stratify the randomization in a study **should be used to stratify the analysis.**

Evaluation of allocation bias in stratified clinical trials

Settings:

- **Multi-component endpoint:**
  - $m = 2$ endpoint components
  - standard normally distributed and uncorrelated
  - evaluated using the stratified Wei-Lachin test.
- **Sample size**: n∈ $\{16,32,64\}$
- **Number of balanced strata**: $K \in \{2,4\}$
- **Allocation bias effect** ~10% of the effect size associated with 80% power
- 10 000 randomization lists



In small trials, over-stratification can amplify type I error inflation under allocation bias.

Keep the number of strata limited and ensure that the number of patients per stratum is not smaller than the number of strata.

Design strategies to mitigate allocation bias in multiple endpoint trials

RANDOMIZATION

Choose less restrictive randomization procedures, e.g. Big Stick Design

→ allows some imbalance
→ may reduce sensitivity to allocation bias

STRATIFICATION

Avoid over-stratification in small trials

More strata
→ fewer patients per stratum
→ greater sensitivity to allocation bias

MULTIPLE ENDPOINTS

Balance information gain against bias sensitivity

More uncorrelated components
→ more information
→ but greater sensitivity to allocation bias

Limit components of the multi-component endpoint

Design strategies to mitigate allocation bias in multiple endpoint trials

RANDOMIZATION

Choose less restrictive randomization procedures, e.g. Big Stick Design

→ allows some
→ may reduce

STRATIFICATI

Avoid over-stratification in small trials

More strata
→ fewer patients per stratum
→ greater sensitivity to allocation bias

MULTIPLE ENDPOINTS

Balance information gain against bias

cation bias

Limit components of the multi-component endpoint

Careful design before trial execution is essential for robust and valid results

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Multi-arm trials

Typical examples of multi-arm trials



Dose finding / dose selection studies



Different treatment durations or regimens



Several experimental treatments vs. one shared control



Platform trials

Unequal allocation in multi-arm trials

"In master protocols evaluating multiple drugs with a shared control arm, sponsors should consider using a randomization scheme that allocates more participants to the control arm than each individual drug arm, as this can increase power for each drug-versus-control comparison for a given total sample size […]"

"It is possible for the randomization ratio to change in the setting of a master protocol. This can occur when products enter or exit a platform trial over time […]. For example, one randomization scheme […] could change the randomization ratio from v2:1:1 (control: drug A: drug B) to v3:1:1:1 (control: drug A : drug B : drug C) when a third drug, drug C, enters a trial." [FDA, 2023]

U.S. Food and Drug Administration. *Master Protocols for Drug and Biological Product Development* (Draft Guidance, Revision 1). CDER and CBER; June 2026.

The given allocation ratio in the guideline is the optimal ratio for a trial conducting pairwise comparisons between multiple active treatment arms and a shared control arm, assuming equal variances for the outcome*.

*Sverdlov, Oleksandr & Ryeznik, Yevgen. (2019). Implementing unequal randomization in clinical trials with heterogeneous treatment costs. Statistics in Medicine. 38. 2905-2927. 10.1002/sim.8160.

Randomization procedures in multi-arm trials

Complete Randomization (CR)

- **Allocation by fair coin toss:** Participants (n) are randomly assigned to the treatment or control group with equal probability ($\frac{1}{2}$ each).



Multi-arm extension

- Every participant is randomized to treatment groups with probabilities equal to the target allocation proportions
  - For example, for 1:1:√2, the *i*-th subject is assigned to treatment A,B,C with probability 0.293/0.293/0.414
- Three exemplary sequences of length *n* = 17:

																	As	Bs	Cs
B	C	A	B	C	A	A	C	A	C	B	A	C	B	B	A	C	6	5	6
B	A	B	B	C	B	C	A	A	C	B	A	C	C	A	B	A	6	6	5
C	C	C	C	A	C	C	A	B	A	C	C	A	A	C	A	C	6	1	10

Permuted Block Randomization (PBR(a))

- **Allocation:** Patients are randomized in blocks of size a; within each block, allocation follows RAR.

Multi-arm extension

- Consider 1:4:4 allocation, and blocks of size 9
- Draw from all eligible 630 permutations with equal probability:

Permutation No	Sequence	Prob
1	ABBBBCCCC	1/630
2	ABBBCCCCB	1/630
…	…	…
630	CCCCBBBBA	1/630

What about square root allocation?

Target ratio	Integer approximation	Block size	Approximation
1 : 1 : √2	5 : 5 : 7	17	√2 ≈ 1.4
1 : 1 : √2	2 : 2 : 3	7	√2 ≈ 1.5

- Problem: Large imbalances possible due to long block sizes. For the first approximation we might get after 24 patients:

BBAABAAABBCCCCCCC CCCCCCC

Mass weighted urn design

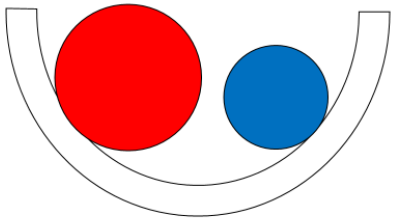
The mass weighted urn design (MWUD) can exactly represent any target allocation without rounding to integer block ratios.

Target allocation ratio ρ = (2/3, 1/3)

Total urn mass α = 3

Initial weights w_0 = (2, 1)

Allocation Step	Treatment	Weights A/B	Probability of getting A
1		2/1	2/3=0.67
2			
3			
4			
5			
6			
7			
8			



Once the blue ball is drawn, increase the red ball's weight by 2/3 and reduce the blue ball's weight by 2/3

Once the red ball is drawn, increase the blue ball's weight by 1/3 and reduce the red ball's weight by 1/3

Zhao W. Mass weighted urn design—a new randomization algorithm for unequal allocations. Contemp Clin Trials. 2015;43:209-216

Mass weighted urn design

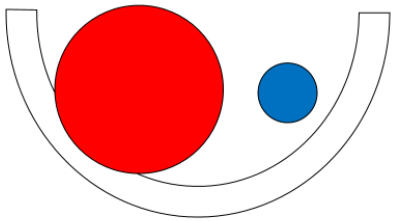
The mass weighted urn design (MWUD) can exactly represent any target allocation without rounding to integer block ratios.

Allocation Step	Treatment	Weights A/B	Probability of getting A
1	B	2/1	2/3=0.67
2		2.67/0.33	2.67/3=0.89
3			
4			
5			
6			
7			
8			

Target allocation ratio $\rho = (2/3, 1/3)$

Total urn mass $\alpha = 3$

Initial weights $w_0 = (2, 1)$



Once the blue ball is drawn, increase the red ball's weight by 2/3 and reduce the blue ball's weight by 2/3

Mass weighted urn design

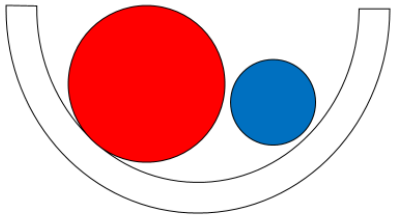
The mass weighted urn design (MWUD) can exactly represent any target allocation without rounding to integer block ratios.

Allocation Step	Treatment	Weights A/B	Probability of getting A
1	B	2/1	2/3=0.67
2	A	2.67/0.33	2.67/3=0.89
3		2.33/0.67	2.33/3=0.78
4			
5			
6			
7			
8			

Target allocation ratio $\rho = (2/3, 1/3)$

Total urn mass $\alpha = 3$

Initial weights $w_0 = (2, 1)$



Once the red ball is drawn, increase the blue ball's weight by 1/3 and reduce the red ball's weight by 1/3

Mass weighted urn design

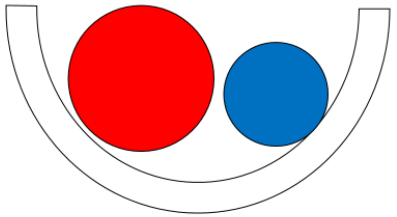
The mass weighted urn design (MWUD) can exactly represent any target allocation without rounding to integer block ratios.

Allocation Step	Treatment	Weights A/B	Probability of getting A
1	B	2/1	2/3=0.67
2	A	2.67/0.33	2.67/3=0.89
3	A	2.33/0.67	2.33/3=0.78
4		2/1	2/3=0.67
5			
6			
7			
8			

Target allocation ratio $\rho = (2/3, 1/3)$

Total urn mass $\alpha = 3$

Initial weights $w_0 = (2, 1)$



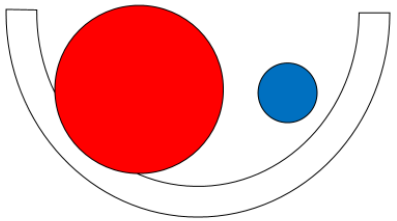
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4	B	2/1	2/3=0.67
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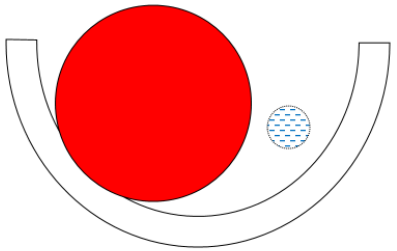
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Target allocation ratio ρ = (2/3, 1/3)

Total urn mass α = 3

Initial weights w_0 = (2, 1)

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3	A	2.33/0.67	2.33/3=0.78
4	B	2/1	2/3=0.67
5	B	2.67/0.33	2.67/3=0.89
6		3.33/**-0.33!**	3.33/3.33=1
7			
8			



Once the blue ball is drawn, increase the red ball's weight by 2/3 and reduce the blue ball's weight by 2/3

It might be possible that a ball receives a negative mass – in such a case, its conditional allocation probability is set to 0

Mass weighted urn design

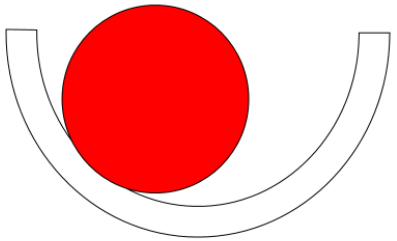
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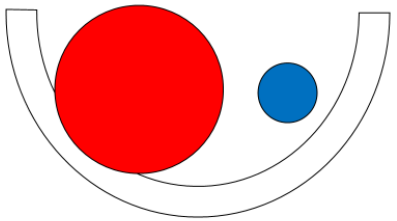
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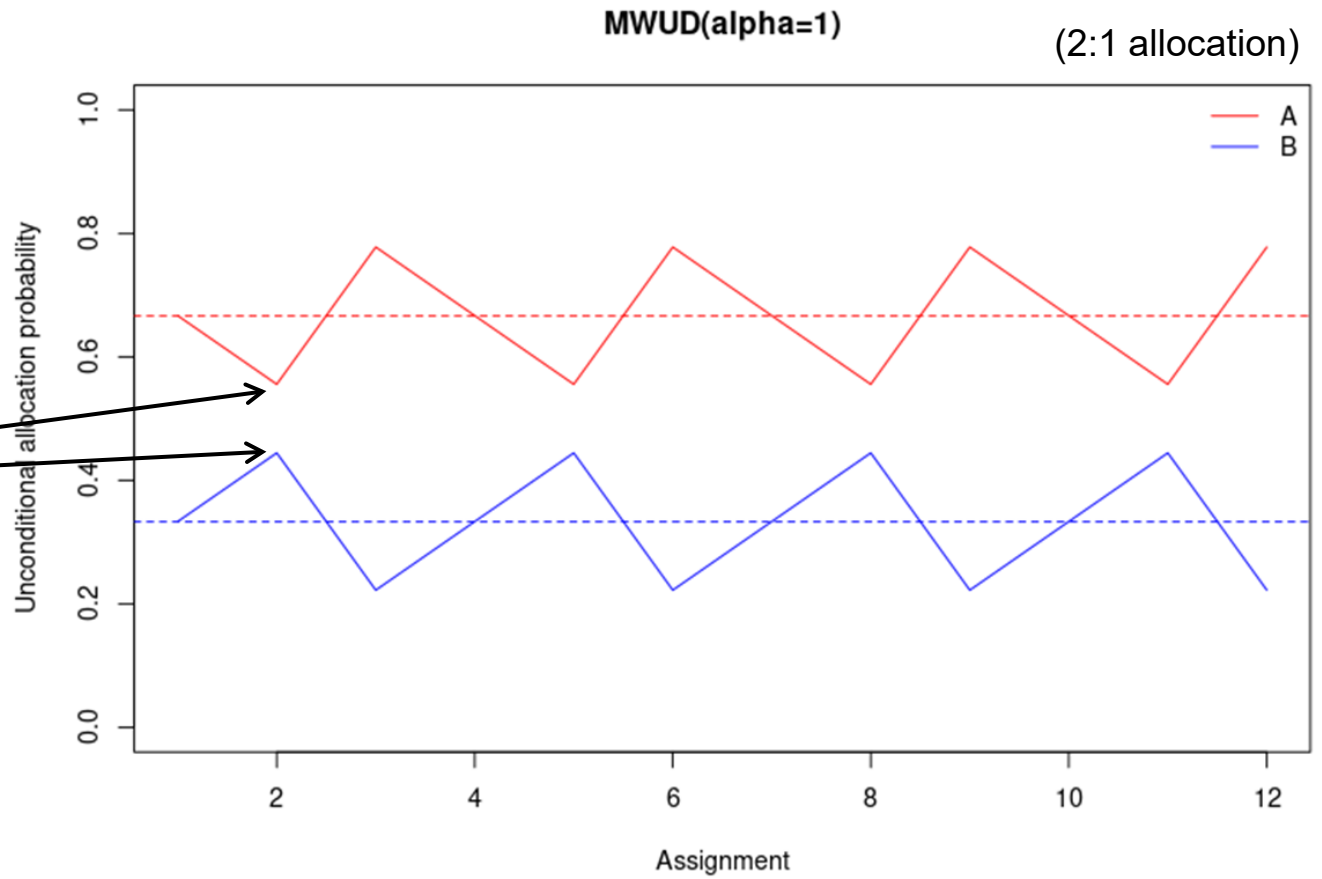


Once the red ball is drawn, increase the blue ball's weight by 1/3 and reduce the red ball's weight by 1/3

Important in multi-arm trials: Allocation ratio preserving (ARP) property

- The **unconditional allocation ratio** for a given participant in the sequence is the allocation ratio in which the participant will be randomized without conditioning on the current treatment arm totals
- An allocation ratio preserving procedure keeps this ratio identical at every allocation step

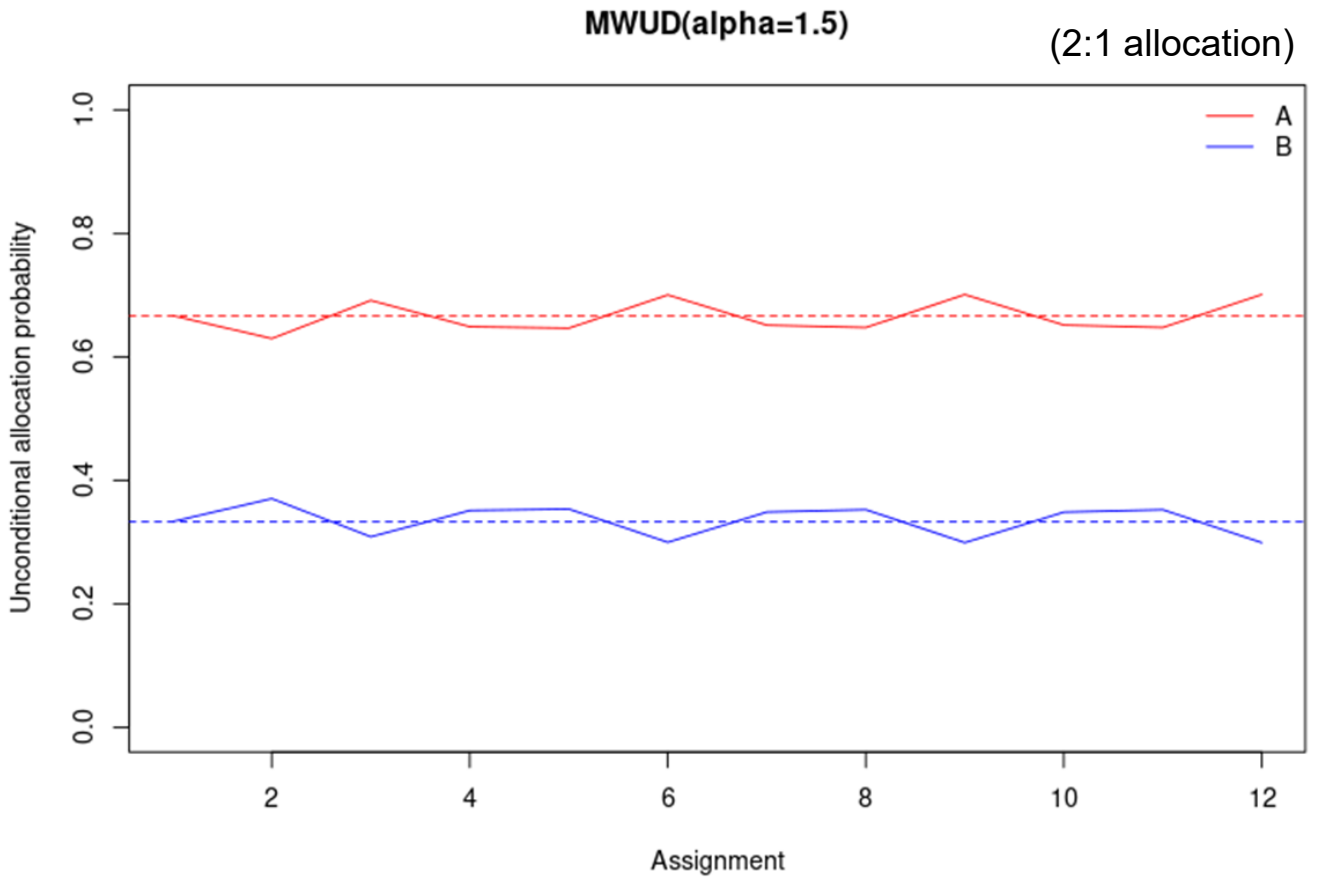
If unconditional allocation ratio is not kept allocation bias can arise (even in double blind trials)!



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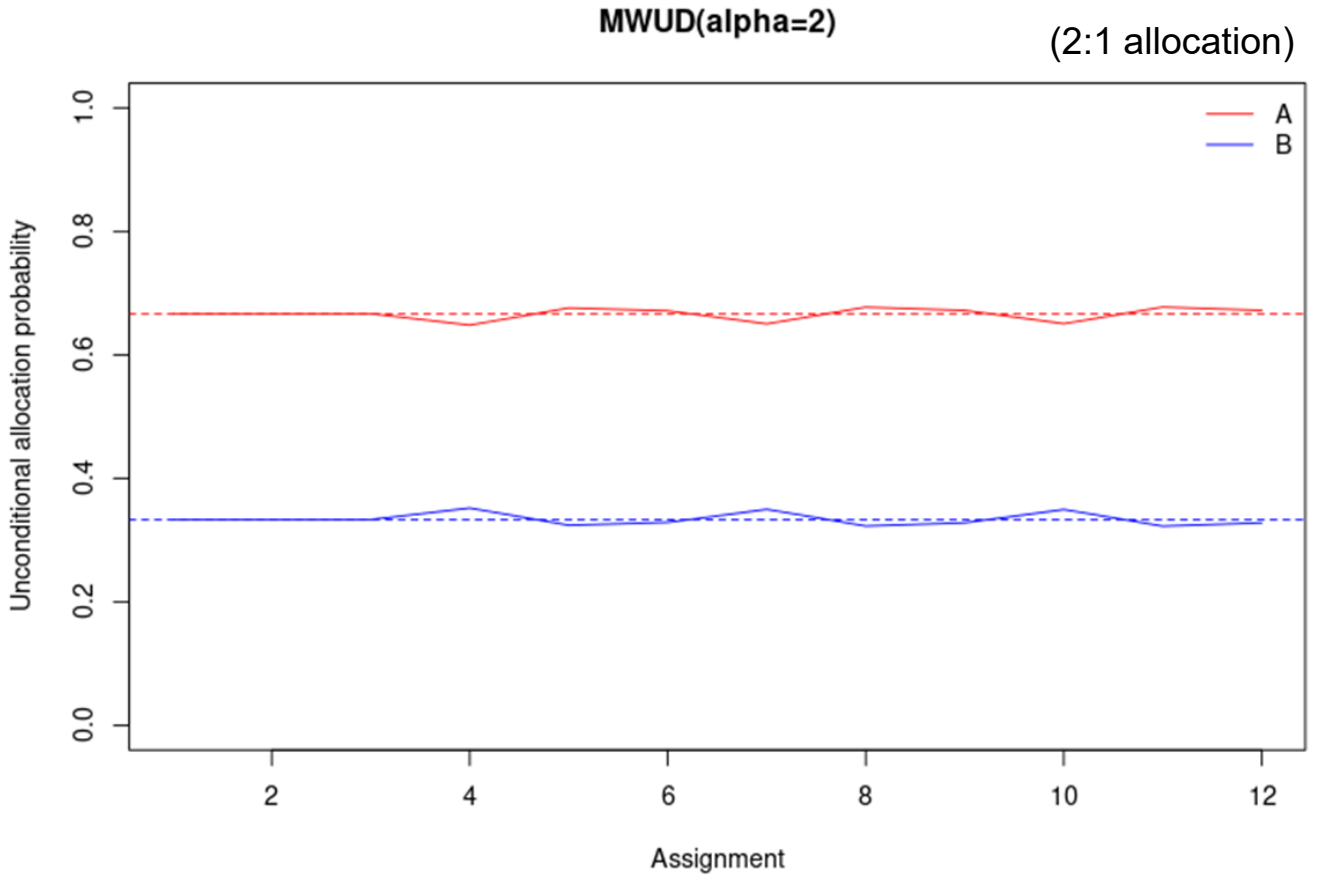
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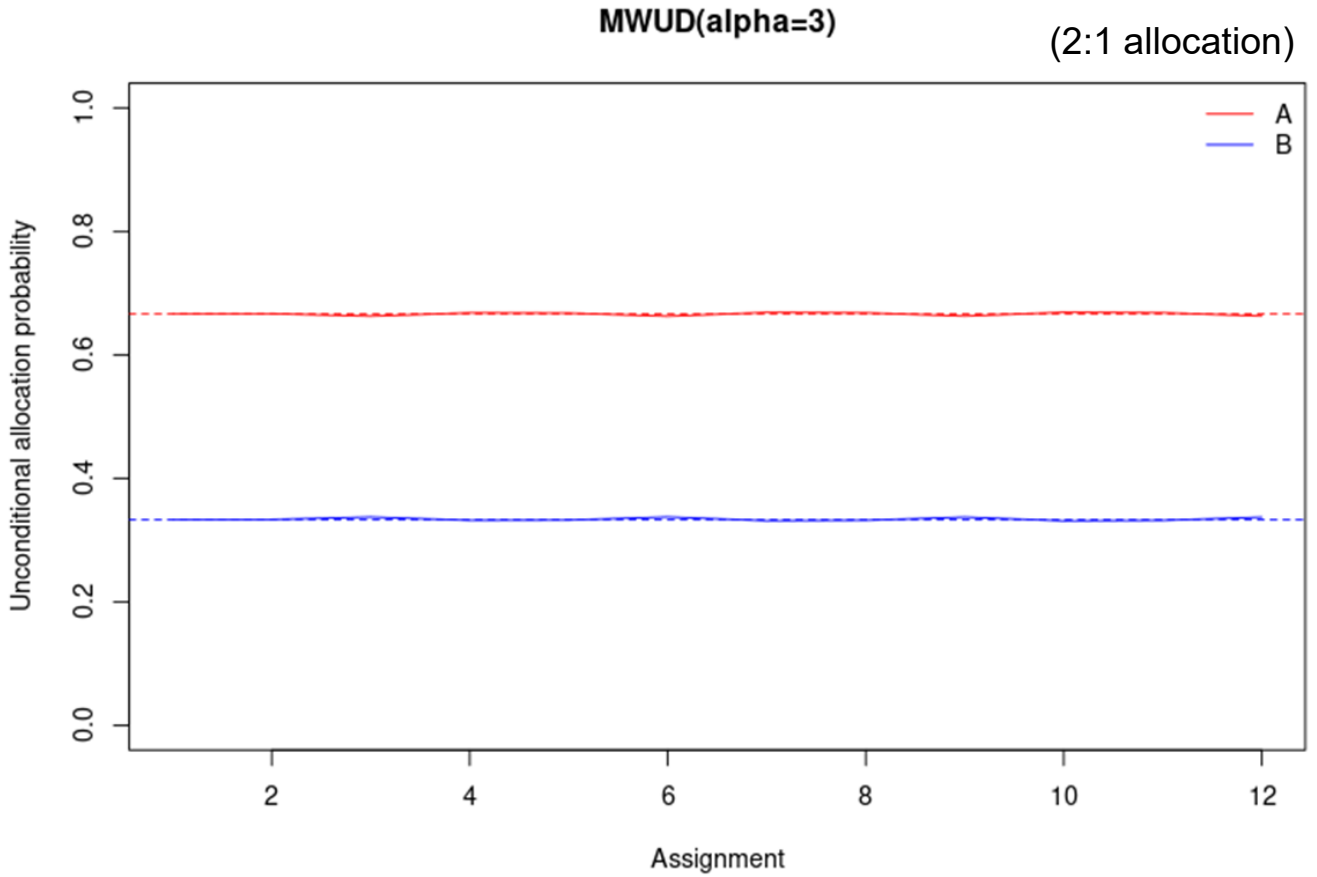
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If unconditional allocation ratio is not kept allocation bias can arise (even in double blind trials)!



Important in multi-arm trials: Allocation ratio preserving (ARP) property

- Complete randomization and permuted block randomization preserve the allocation ratio at every allocation step.
- Mass weighted urn design allows for any target allocation ratios, but does not preserve the allocation ratio for every parameter α.
- Other procedures exist that keep the ARP exactly, like brick tunnel randomization[1] but at the cost of other trade-offs like complexity.
- When using unequal allocation, check the ARP property before choosing a procedure!

[1] Kuznetsova OM, Tymofyeyev Y. Brick tunnel randomization for unequal allocation to two or more treatment groups. Stat Med. 2011;30(8):812–824.

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Randomization in Platform Trials

What is a platform trial?

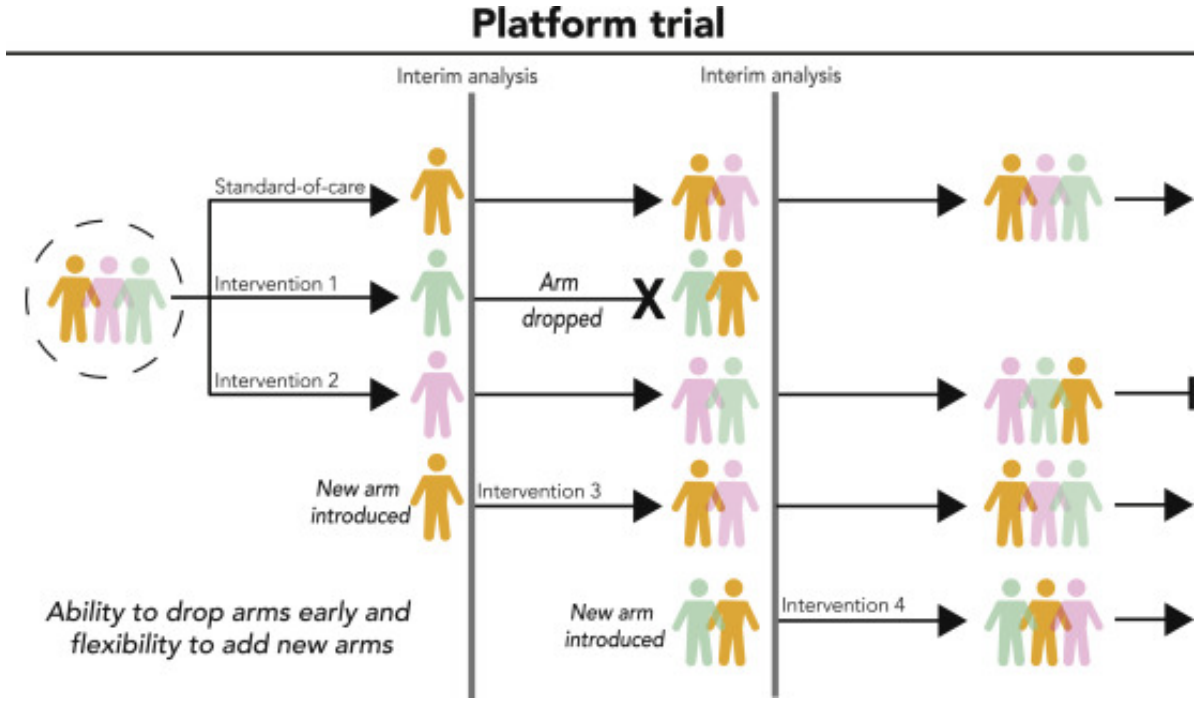
A platform trial investigates

"multiple targeted therapies in the context of a single disease in a perpetual manner, with therapies allowed to enter or leave the platform on the basis of a decision algorithm."

Adaptive Platform Trials Coalition, *Nat Rev Drug Discov* 18, 797–807 (2019)

Key Features

- One master protocol
- Shared control arm
- Adaptive design



Source: Park et al. (2020), Journal of Clinical Epidemiology

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Why platform trials?

Platform trials share infrastructure and a common control arm across multiple experimental arms, enabling new arms to be added or dropped based on interim evidence.

Advantages:
→ Operational efficiency via shared infrastructure
→ More efficient use of the patient pool via a common control group



But: Operational flexibility introduces bias risks and introduces additional complexity

How to randomize in a platform trial with dynamically changing allocation ratios and frequent interim analyses?

FDA Draft Guidance on Master Protocols

III. CONSIDERATIONS FOR DESIGN AND ANALYSIS

This section discusses important considerations for the design and analysis of master protocols, with a focus on randomized trials that are intended to contribute to a demonstration of safety and substantial evidence of effectiveness.

A. Randomization

FDA recommends randomization of participants to the drug (or to one of the drugs) being evaluated or to a control arm to remove systematic imbalances between treatment arms in both measured and unmeasured prognostic factors and to ensure reliable inference on drug safety and effectiveness.

In master protocols evaluating multiple drugs with a shared control arm, sponsors should consider using a randomization scheme that allocates more participants to the control arm than each individual drug arm, as this can increase power for each drug-versus-control comparison for a given total sample size (Chandereng et al. 2020 and Appendix: section A). Note that although the randomization ratio that optimizes power involves greater-than-equal allocation to the control arm, the probability that an individual participant entering the trial will be assigned to control is less than in a typical two-arm controlled trial with 1:1 randomization. This disproportionate randomization also reduces the risk of a poorly or highly performing control arm leading to multiple correlated erroneous findings (see section III.H).

The choice of a randomization scheme can be informed by multiple factors, including efficiency

Randomization is a major theme of the draft guidance

U.S. Food and Drug Administration. *Master Protocols for Drug and Biological Product Development* (Draft Guidance, Revision 1). CDER and CBER; June 2026.

Randomization challenges in platform trials

Randomization in platform trials has to deal with:

- Arms added and closed while the trial runs
- Frequent updates of the allocation ratio
- Unequal allocation
- Potentially response-adaptive randomization
- Eligibility differing between arms
- Infeasibility of obtaining informed consent for all treatments for all arms for each patient
- Limited drug supply at small centres

Discussed in detail in


REVIEW Open Access

Randomization in the age of platform trials: unexplored challenges and some potential solutions

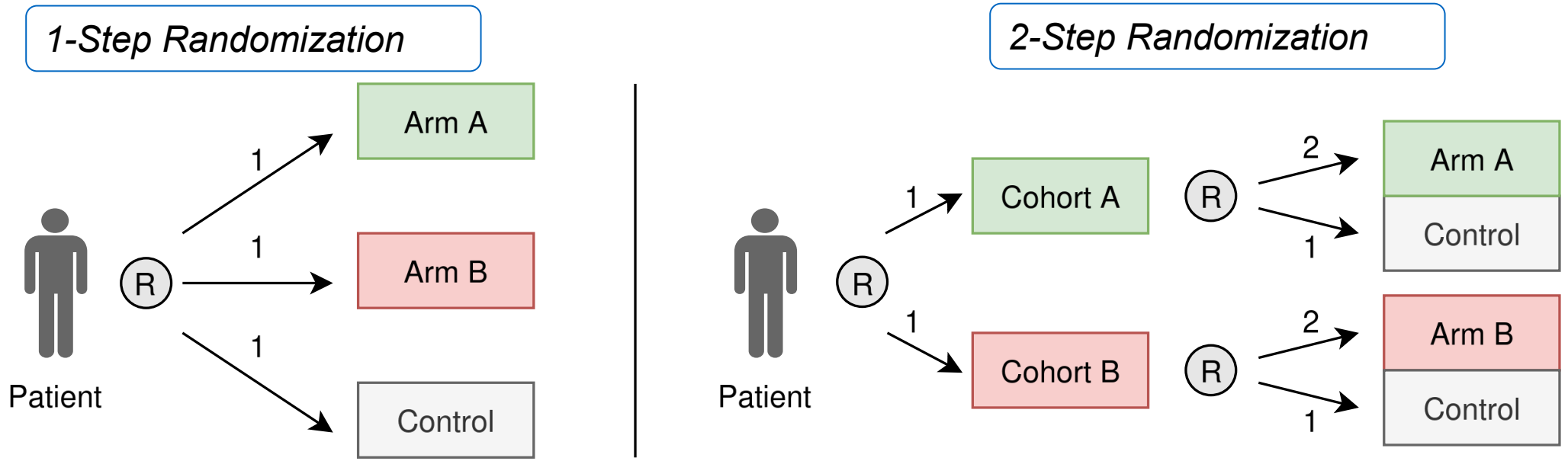
Olga Kuznetsova[1], Jennifer Ross[2], Daniel Bodden[3], Freda Cooner[4], Jonathan Chipman[5], Peter Jacko[6], Johannes Krisam[7], Yuqun Abigail Luo[4], Tobias Mielke[8], David S. Robertson[9], Yevgen Ryeznik[10*], Sofia S. Villar[9], Wenle Zhao[11] and Oleksandr Sverdlov[12*]

Kuznetsova et al. (2025). Randomization in the age of platform trials: unexplored challenges and some potential solutions. BMC Medical Research Methodology 25:268.

Platform trials: 1-step vs. 2-step randomization

Problem: obtaining informed consent for all treatments from every patient may not be feasible

Solution: 2-step randomization. First to cohort, then to treatment or control within the cohort



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Chronological bias & allocation bias in platform trials

2-step randomization

2-step randomization is described for platform trials (FDA 2023), what happens if the investigator knows the first step of the procedure

→ **Allocation bias?**

Long enrolment periods

Platform trials often run over long enrolment periods (EMA 2022), so the participant population and standard of care can drift over time

→ **Chronological bias?**

Both questions in our talk at this conference

Multiple treatment arms, multiple biases? Allocation and chronological biases in rare disease platform trials

D Bodden, N Bruder, F König, R-D Hilgers · 30 Sep 2026, 15:00–15:15 · Oral session: Platform and Cluster Trial Designs

References

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Block IV — Recommendations & take-home messages

- Key lessons learned and further topics
- Open-research question and working groups

Time	Session	Topics
8:30-8:40 (10 min)	Welcome & workshop overview	• Overview of the workshop topics and flow
8:40-9:40 (60 min)	I Block I — General introduction to randomization	• History and evolution of randomization • Randomization, allocation concealment, and blinding • Randomization procedures and their key characteristics • Regulatory perspectives on randomization
9:40-10:30 (50 min)	II Block II — Bias & transparent reporting	• (Allocation) Bias in randomized clinical trials, • Transparent reporting: what to report in the trial protocol and publication
10:30-11:00 (30 min)	Coffee Break	
11:00-12:15 (75 min)	III Block III — Randomization in complex trial designs	• Randomization in small-sample group-sequential designs • (Stratified) randomization and allocation bias in multi-endpoint trials • Randomization in multi-arm and platform trials
12:15-12:30 (15 min)	IV Block IV — Recommendations & take-home messages	• Key lessons learned and further topics • Open-research question and working groups

What randomization does — and does not do

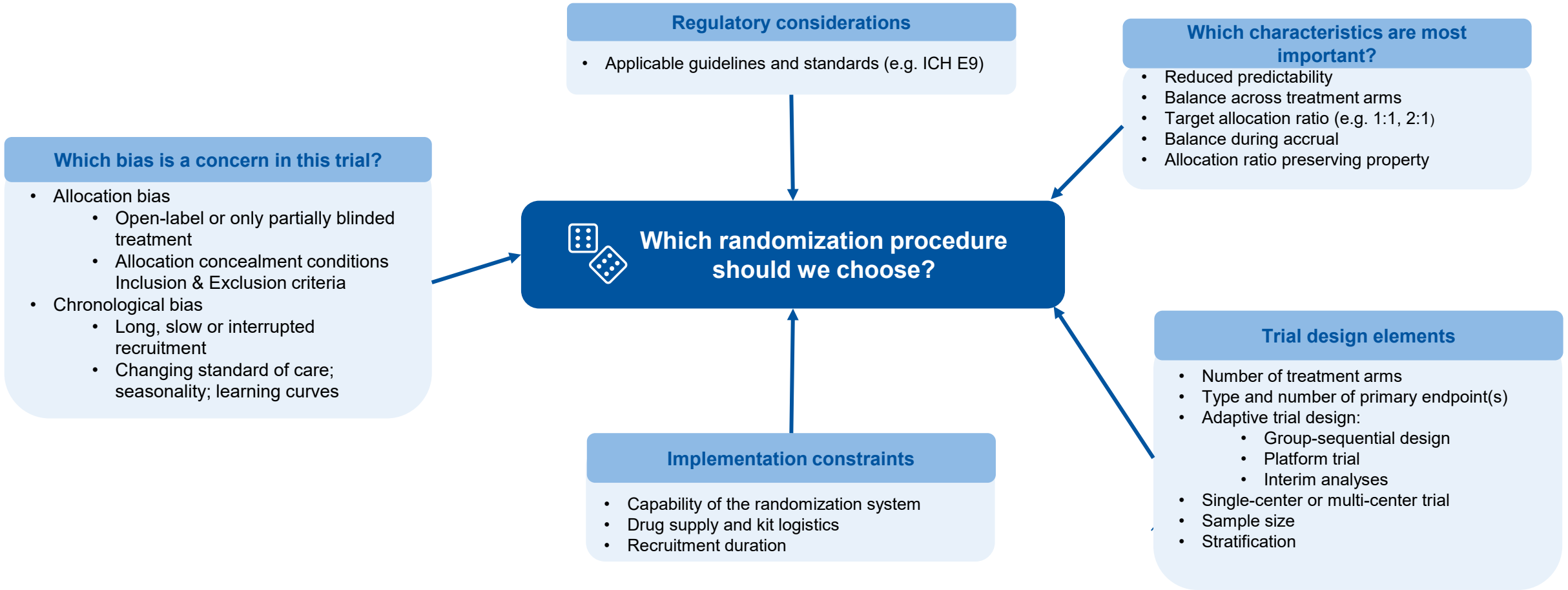
Randomization…

- reduces bias and supports valid statistical inference
- can perform differently depending on the randomization procedure and study design
- balances known and unknown prognostic factors on average

Randomization does not…

- guarantee perfectly identical groups
- prevent allocation bias and other types of bias completely by itself
- remove the need for careful analysis and reporting

What drives the selection of a specific randomization procedure?



There is no one-size-fits-all randomization procedure.

Randomization Working Group

The **Randomization Working Group** is a group of statisticians from industry, academia and regulatory, working to promote the use of novel randomization methods and advance the scientific understanding of these methods in the global community. We are a Special Interest Group of PSI, members of the ASA Biopharmaceutical Section, and the IDSWG within DahShu.

Some topics are:

- Methodological research on statistical properties of randomization methods
- Randomization-based inference
- Development of software tools

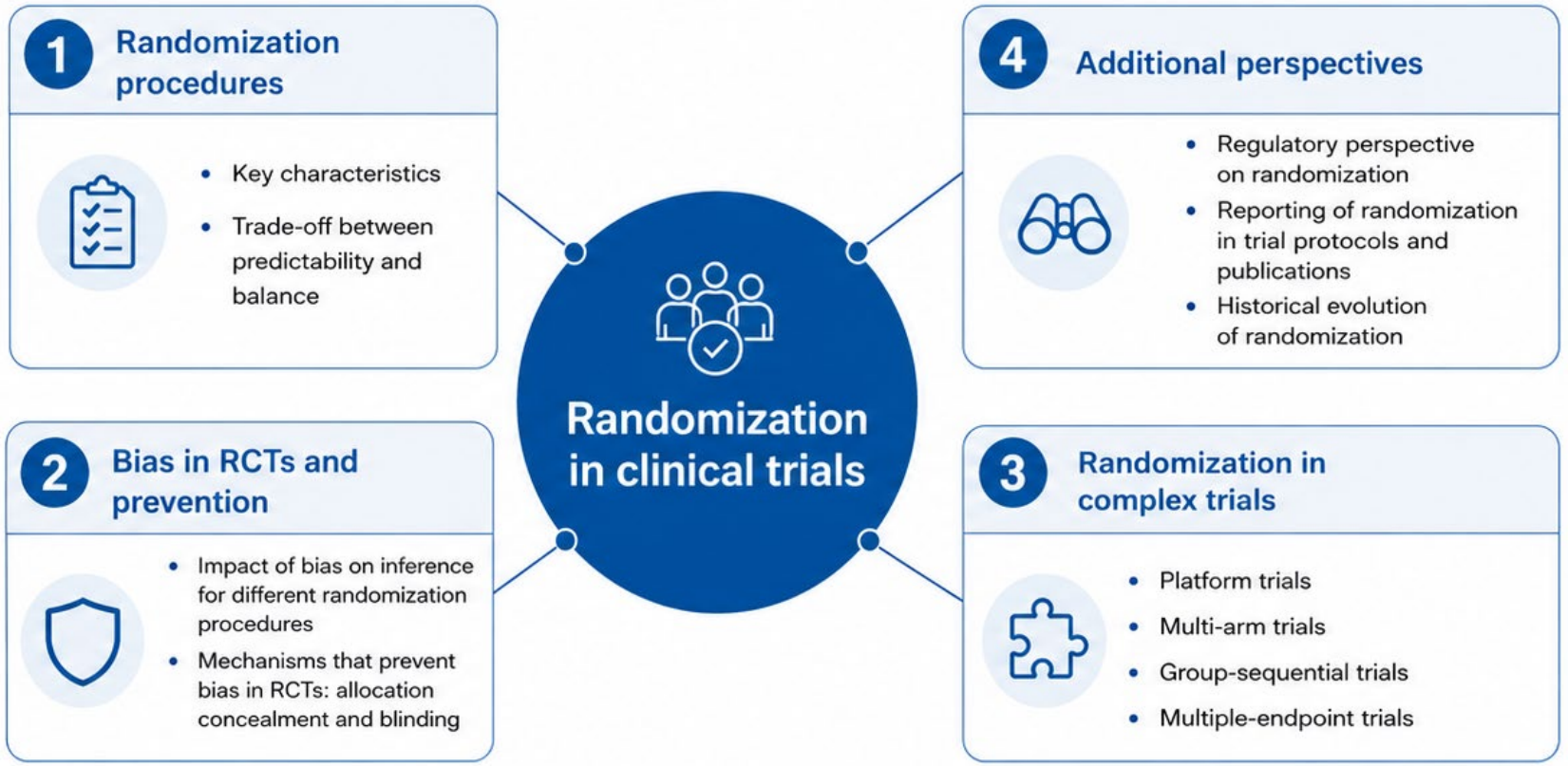
Special thanks to Diane Uschner, Johannes Krisam, Alex Sverdlov for their help preparing this workshop.

randomization working group

rwg

Learn more about our work: https://randomization-wg.org/
.. or join our Working Group on LinkedIn!

What we covered today…



Further topics
(references in the appendix)

- Randomization-based inference
- Covariate-adaptive randomization
- Response-adaptive randomization

Choose the randomization strategy with the whole trial in mind.

Thank you for your attention!

Properties of the randomization procedures

Abbr.	Procedure	Mechanism	Properties
CR	Complete Randomization	Allocation by a fair coin toss.	• Each allocation is equally likely, making the next assignment unpredictable. • Randomization can lead to unequal group sizes and, in rare cases, substantial imbalances
EBC(p)	Efron's Biased Coin	Biased coin toss (p) that favors the under-represented group.	• With p = 2/3, the probability of achieving perfect balance is 1/2 for even n and approaches 3/4 for odd n as n becomes large
RAR	Random Allocation Rule	Exactly N/2 patients are assigned to each group.	• Once $n/2$ patients have been assigned to one group, all remaining allocations are deterministic and fully predictable • No imbalance in group sizes occurs at the end of the randomization process
PBR(a)	Permuted Block Rand.	Patients are randomized in blocks of size a; within each block, allocation follows RAR.	• Possible differences in group sizes during the randomization process is a/2 • Each completed block contains equal numbers of treatment and control assignments, ensuring no imbalance at the end but higher predictability
rPBR(a)	Random permuted Block Rand.	The block length is randomly selected from a predefined set of possible block lengths rather than being fixed.	• Variable block sizes reduces predictability • Each block is filled except possibly the last; if the trial ends with a complete final block, the overall treatment imbalance is zero
BSD(b)	Big Stick Design	CR until imbalance b is reached, then deterministic assignment to the under-represented group.	• Unequal treatment group sizes may remain at the end of randomization • Imbalance is controlled through the maximum tolerated imbalance b
Chen(p,b)	Chen's Urn Design	Biased coin (p) that favors the under-represented group, allowing for imbalance within a limit b.	• Unequal treatment group sizes may remain at the end of randomization • Imbalance is controlled through the maximum tolerated imbalance b

Rosenberger WF, Lachin JM. Randomization in Clinical Trials: Theory and Practice. New York: Wiley; 2002.

How is randomization done in practice?

Industry / large (complex) clinical trials

- Delegated to validated interactive Response Technology system (also couples' randomization with drug-supply logistics) – see Kuznetsova et al. (2025), BMC Med Res Methodol 25:268

Academic / free tools

- randomizeR (R): generate and assess randomization procedures

- Sealed Envelope (web-based): free for non-commercial studies (<= 50 randomization lists)

References

- Kuznetsova O, Ross J, Bodden D, Cooner F, Chipman J, Jacko P, et al. Randomization in the age of platform trials: unexplored challenges and some potential solutions. *BMC Med Res Methodol*. 2025;25(1):268. doi:10.1186/s12874-025-02693-0 PMID: 41275089. PMCID: PMC12670786.
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Randomization-based inference

> "The act of randomization itself provides a basis for inference when random sampling is not conducted from a population model."
>
> Rosenberger WF. Randomization: The forgotten component of the randomized clinical trial. *Stat Med.* 2019;38(1):23–30. doi:10.1002/sim.7901.

Main Idea

Null hypothesis: Treatment assignment E vs. C has no effect on the responses of the n randomized patients.

1. *Calculate the observed test statistic*
2. *Consider all possible treatment assignments permitted by the randomization procedure*
3. *Recalculate the statistic for each assignment*
4. *Sum the probabilities of assignments producing a result ≥ as extreme as observed*
5. *This probability = p-value*

Key message: Randomization tests assumption-free with respect to the response distribution, but the validity of the test depends explicitly on the randomization procedure used

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- Onghena, P. (2018). Randomization tests or permutation tests? A historical and terminological clarification. In V. Berger (Ed.), *Randomization, masking, and allocation concealment* (pp. 209-227). Boca Raton/FL: Chapman & Hall/CRC Press.

Response adaptive randomization (RAR)

> *"In a RAR procedure, the allocation probabilities that define the sampling strategy are adapted based on the past treatment allocations and response data."*
>
> Robertson, Lee, López-Kolkovska & Villar (2023), *Statistical Science* 38(2):185–208, 10.1214/22-STS865

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Covariate-adaptive randomization and minimization

> *"Minimisation is based on a different principle from randomisation. The first participant is allocated a treatment at random. For each subsequent participant we determine which treatment would lead to better balance between the groups in the variables of interest."*
>
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>
> *"by covariate-adaptive randomization, we mean randomization schemes that first stratify according to baseline covariates and then assign treatment status so as to achieve 'balance' within each stratum"*
>
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